

数学物理方程



5 Green 函数法

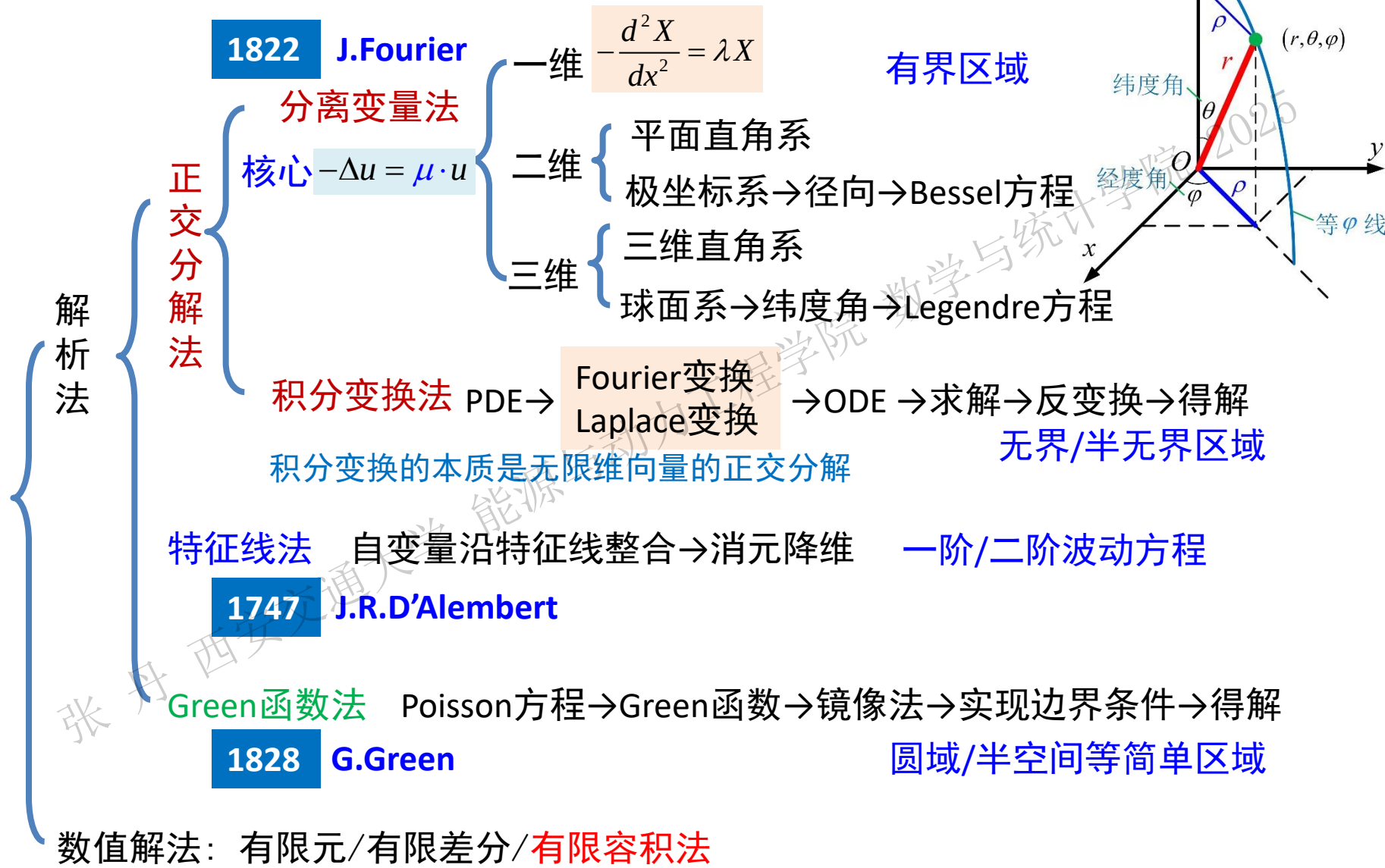
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能源与动力工程学院

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动力工程多相流国家重点实验室

PDE的解法



Green函数法



G.Green
1793-1841
British Math.
Physical Scientist

Green
函数法

波动方程

导热方程

Poisson方程

$$\begin{cases} -\Delta G = \delta(x, y, z), & (x, y, z) \in \Omega \\ G(x, y, z) = 0, & (x, y, z) \in \partial\Omega \end{cases}$$

Green函数是该定解问题的解

- 1. Poisson方程描述变量稳态分布
- 2. Dirichlet问题→给出了区域形状+边界条件+自由项→能否得出内部变量分布显式解
- 3. 圆域Laplace方程→Poisson公式证明该表示是有可能的→Green函数法则给出更通用的表达式

Dirichlet问题 第一类边界

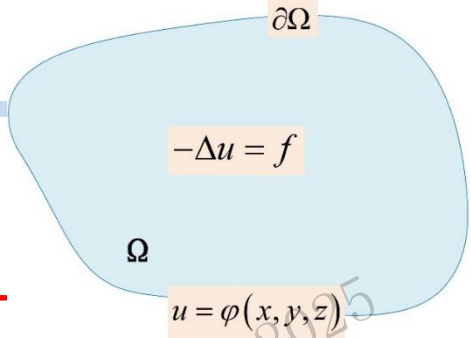
$$\begin{cases} \Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ u(x, y, z) = \varphi(x, y, z), & (x, y, z) \in \partial\Omega \end{cases}$$
$$u(\xi, \eta, \zeta) = - \iint_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV$$

Neumann问题 第二类边界

$$\begin{cases} \Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ \frac{\partial u}{\partial n}(x, y, z) = \varphi(x, y, z), & (x, y, z) \in \partial\Omega \end{cases}$$

圆域上调和函数的Poisson公式

$$u(\rho, \theta) = \frac{1}{2\pi} \int_0^{2\pi} \frac{(a^2 - \rho^2)}{a^2 + \rho^2 - 2a\rho \cos(\theta - \tau)} \varphi(\tau) d\tau$$



J. Dirichlet
1805-1859
German Math.
Physical Scientist



C. Neumann
1832-1925
German Math.

数学物理方程

1. 数学建模和基本原理介绍

2. 分离变量法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

5.1 Green公式

5.2 Laplace方程基本解

5.3 Green函数

5.4 半空间/圆域上的Dirichlet问题

5.1 Green公式

预备知识 梯度/方向导数/散度

Hamilton算子

$$\nabla = \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k}$$

W. Hamilton
1805-1865
Ireland Math.
Physical Scientist



标量 $f \rightarrow \nabla \rightarrow$ 矢量 $\rightarrow$ 梯度

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

矢量 $\mathbf{u} \rightarrow \nabla \rightarrow$ 标量 $\rightarrow$ 散度

点积

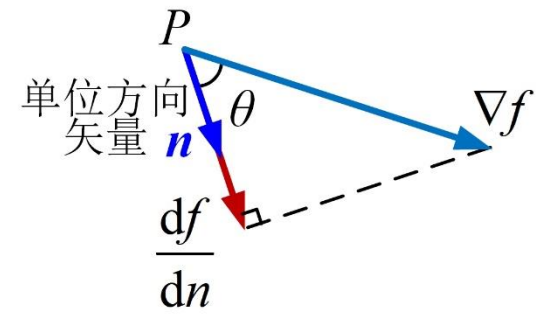
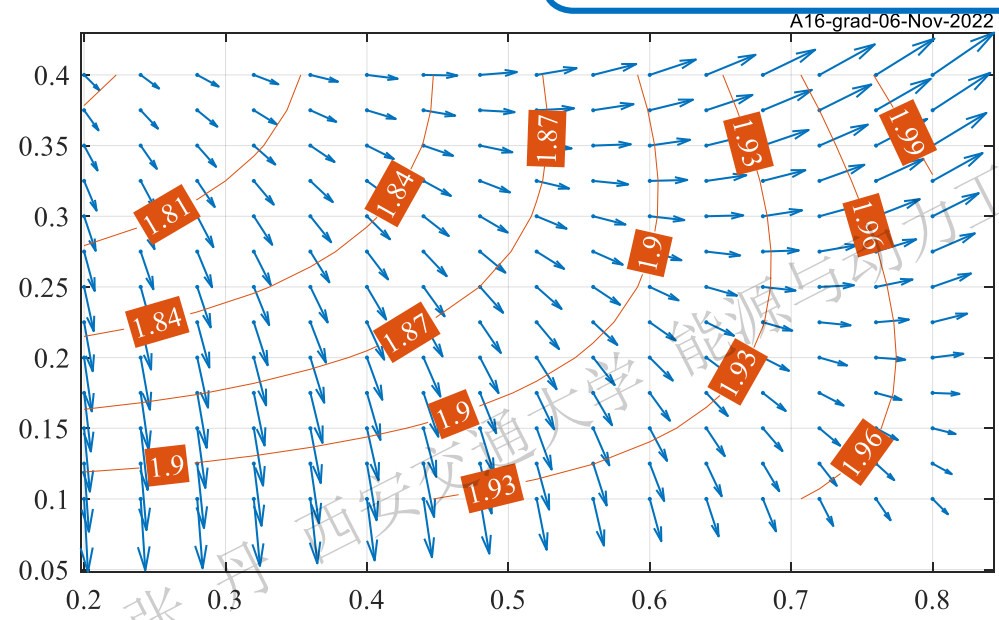
$$\nabla \cdot \mathbf{u} = \nabla \cdot (u_x, u_y, u_z) = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}$$



P.Laplace
1749-1827
French Math.
Physical Scientist

Laplace等式

$$\nabla \cdot \nabla f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \Delta f$$



大小 $\rightarrow$ 标量最大变化率

方向 { 标量等值面的法线
给定点处标量最大
变化率的方向

标量场沿指定方向 $\mathbf{n}$ 的变化率

$$\nabla f \cdot \mathbf{n} = |\nabla f| \cdot 1 \cdot \cos \theta = \frac{df}{dn} \quad \text{方向导数}$$

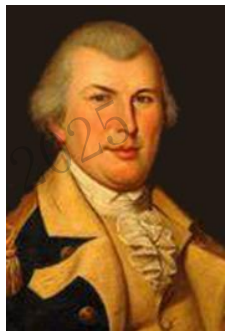
5.1 Green公式

Gauss公式→Green公式

用边界表示内部→推导Green公式→探寻 $\Delta u, u, \frac{\partial u}{\partial n}$ 固有关系



J. Gauss
1777-1855
German Math.
Physical Scientist



G.Green
1793-1841
British Math.

Green第1公式

$$\iiint_{\Omega} u \Delta v dV = \iint_{\partial\Omega} u \frac{\partial v}{\partial n} ds - \iiint_{\Omega} \nabla v \cdot \nabla u dV$$

Green第2公式

$$\iiint_{\Omega} (u \Delta v - v \Delta u) dV = \iint_{\partial\Omega} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds$$

$$\iiint_{\Omega} \nabla \cdot F dV = \iint_{\partial\Omega} F \cdot n ds$$

关联了内部/边界

基本解

三维 $\Gamma(P, P_0) = \frac{1}{4\pi r_{P_0 P}} = \frac{1}{4\pi} \cdot \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}}$

二维 $\Gamma(P, P_0) = \frac{1}{2\pi} \ln \frac{1}{r_{P_0 P}} = \frac{1}{2\pi} \ln \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2}}$

Green第3公式

$$u(\xi, \eta, \zeta) = \iint_{\partial\Omega} \left(\Gamma \frac{\partial u}{\partial n} - u \frac{\partial \Gamma}{\partial n} \right) ds - \iiint_{\Omega} \Gamma \Delta u dV$$

Green公式是Gauss公式的推广

5.1 Green公式

Green公式 G1/G2公式 函数 $u(x,y,z), v(x,y,z)$ 在给定区域/边界上二阶可导



G.Green
1793-1841
British Math.

令 $F = u \nabla v \rightarrow$ Gauss公式 $\iiint_{\Omega} \nabla \cdot F dV = \iint_{\partial\Omega} F \cdot n ds$

乘积求导 $\iiint_{\Omega} \nabla \cdot (u \nabla v) dV = \iint_{\partial\Omega} u \nabla v \cdot n ds$ 方向导数定义

$$\iiint_{\Omega} u \nabla \cdot (\nabla v) dV + \iiint_{\Omega} \nabla u \cdot \nabla v dV = \iint_{\partial\Omega} u \frac{\partial v}{\partial n} ds$$

$$\iiint_{\Omega} u \Delta v dV + \iiint_{\Omega} \nabla u \cdot \nabla v dV = \iint_{\partial\Omega} u \frac{\partial v}{\partial n} ds$$

Green第1公式 $\iiint_{\Omega} u \Delta v dV = \iint_{\partial\Omega} u \frac{\partial v}{\partial n} ds - \iiint_{\Omega} \nabla u \cdot \nabla v dV$ ①

上述推导中 u, v 地位相同 $\rightarrow$ 位置互换可得

$$\iiint_{\Omega} v \Delta u dV = \iint_{\partial\Omega} v \frac{\partial u}{\partial n} ds - \iiint_{\Omega} \nabla u \cdot \nabla v dV$$
 ②

式 ① ② 相减得

Green第2公式 $\iiint_{\Omega} (u \Delta v - v \Delta u) dV = \iint_{\partial\Omega} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds$

- 1. 消去了梯度乘积项
- 2. 出现了 $\Delta u \rightarrow$ 需进一步关联未知函数 u

5.1 Green公式

Green公式 距离函数

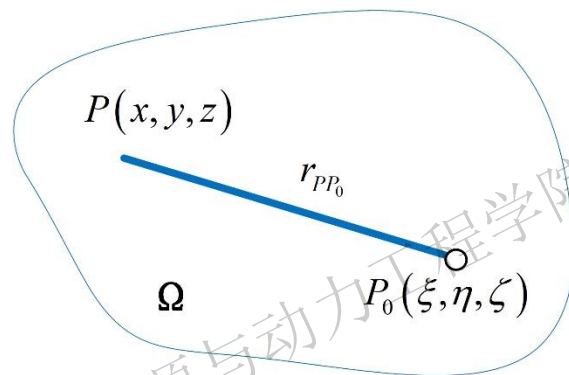
Green第1公式

$$\iiint_{\Omega} u \Delta v dV = \iint_{\partial\Omega} u \frac{\partial v}{\partial n} ds - \iiint_{\Omega} \nabla u \cdot \nabla v dV$$

Green第2公式

$$\iiint_{\Omega} (u \Delta v - v \Delta u) dV = \iint_{\partial\Omega} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds$$

设定点 $P_0(\xi, \eta, \zeta) \in \Omega$, 则该区域中任意点 $P(x, y, z)$ 与 P_0 存在距离函数



ξ /'ksai/
 ζ /'zi:tə/

$$r_{P_0P} = |P(x, y, z) - P_0(\xi, \eta, \zeta)| = \sqrt{(x - \xi)^2 + (y - \eta)^2 + (z - \zeta)^2}$$

1. 距离函数是6元函数 $r_{P_0P} = r_{P_0P}(x, y, z, \xi, \eta, \zeta)$, 两个坐标点的关联函数
2. 坐标 (x, y, z) 与 (ξ, η, ζ) 地位完全平等
3. 距离函数沿坐标系每个方向的求导/积分默认对 (x, y, z) 操作
4. 距离函数 $\rightarrow$ 给定区域上的定积分 $\rightarrow (x, y, z)$ 消失 $\rightarrow$ 未知函数多用 $u(\xi, \eta, \zeta)$ 表示

5.1 Green公式

Green公式

Laplace方程的基本解

Green第1公式

$$\iiint_{\Omega} u \Delta v dV = \iint_{\partial\Omega} u \frac{\partial v}{\partial n} ds - \iiint_{\Omega} \nabla u \cdot \nabla v dV$$

Green第2公式

$$\iiint_{\Omega} (u \Delta v - v \Delta u) dV = \iint_{\partial\Omega} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds$$

距离函数 $r_{P_0P} = |P(x, y, z) - P_0(\xi, \eta, \zeta)| = \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}$

基本解 $\Gamma(P, P_0) = \frac{1}{4\pi r_{P_0P}} = \frac{1}{4\pi} \cdot \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}} = \Gamma(x, y, z)$

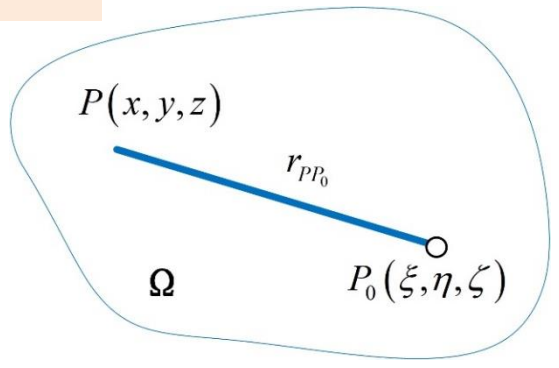
验证基本解满足Laplace方程

$$\begin{aligned} r_x &= \frac{\partial r_{P_0P}}{\partial x} = \frac{1}{2} \left[(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2 \right]^{-\frac{1}{2}} 2(x-\xi) = \frac{x-\xi}{r} \\ \Gamma_x &= \frac{d\Gamma}{dr} \frac{\partial r_{P_0P}}{\partial x} = -\frac{1}{4\pi} r^{-2} r_x = -\frac{1}{4\pi} r^{-2} \frac{x-\xi}{r} = -\frac{1}{4\pi} r^{-3} (x-\xi) \\ \Gamma_{xx} &= \frac{3}{4\pi} r^{-4} (x-\xi) \frac{x-\xi}{r} - \frac{1}{4\pi} r^{-3} = \frac{2(x-\xi)^2 - (y-\eta)^2 - (z-\zeta)^2}{4\pi r^5} \\ \Gamma_{yy} &= \frac{2(y-\eta)^2 - (x-\xi)^2 - (z-\zeta)^2}{4\pi r^5} \\ \Gamma_{zz} &= \frac{2(z-\zeta)^2 - (x-\xi)^2 - (y-\eta)^2}{4\pi r^5} \end{aligned}$$

同理对 y, z 求导有

$$\Gamma_{xx} + \Gamma_{yy} + \Gamma_{zz} = \Delta \Gamma = 0$$

空间 Ω 除定点 $P_0(\xi, \eta, \zeta)$ 外, $\Gamma(x, y, z)$ 均满足Laplace方程



5.1 Green公式

Green公式 G3公式

Green第1公式
$$\iiint_{\Omega} u \Delta v dV = \iint_{\partial \Omega} u \frac{\partial v}{\partial n} ds - \iiint_{\Omega} \nabla u \cdot \nabla v dV$$

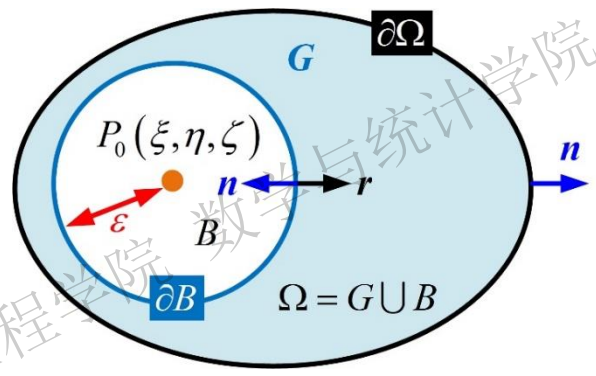
Green第2公式
$$\iiint_{\Omega} (u \Delta v - v \Delta u) dV = \iint_{\partial \Omega} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds$$



G.Green
1793-1841
British Math.

Green第2公式 → 基本解 → Green第3公式

去除奇点 以定点 $P_0(\xi, \eta, \zeta)$ 为球心取球形小邻域 $\bar{B} = B \cup \partial B$, 其余空间记为 $G = \Omega \setminus \bar{B}$



在区域 $G = \Omega \setminus \bar{B}$ 内运用Green第2公式, 并取

基于 P_0 的基本解
$$v = \Gamma(P, P_0) = \frac{1}{4\pi} \cdot \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}}$$

$$\begin{aligned} \iiint_G (u \Delta \Gamma - \Gamma \Delta u) dV &= \iint_{\partial G} \left(u \frac{\partial \Gamma}{\partial n} - \Gamma \frac{\partial u}{\partial n} \right) ds \\ &= 0 - \iint_G \Gamma \Delta u dV = \iint_{\partial \Omega} \left(u \frac{\partial \Gamma}{\partial n} - \Gamma \frac{\partial u}{\partial n} \right) ds + \iint_{\partial B} \left(u \frac{\partial \Gamma}{\partial n} - \Gamma \frac{\partial u}{\partial n} \right) ds \end{aligned}$$

2个边界 ∂B 与 $\partial \Omega$

① ②

在球形邻域边界 ∂B 上化简这两项

5.1 Green公式

Green公式 **G3公式**

Green第1公式
$$\iiint_{\Omega} u \Delta v dV = \iint_{\partial\Omega} u \frac{\partial v}{\partial n} ds - \iiint_{\Omega} \nabla u \cdot \nabla v dV$$

Green第2公式
$$\iiint_{\Omega} (u \Delta v - v \Delta u) dV = \iint_{\partial\Omega} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds$$



$$-\iiint_G \Gamma \Delta u dV = \iint_{\partial\Omega} \left(u \frac{\partial \Gamma}{\partial n} - \Gamma \frac{\partial u}{\partial n} \right) ds + \iint_{\partial B} \left(u \frac{\partial \Gamma}{\partial n} - \Gamma \frac{\partial u}{\partial n} \right) ds$$

1 **2**

在球形小邻域 $\bar{B}$ 的外边界上

外法向量为 $n = -r$

$$\frac{\partial \Gamma}{\partial n} = \frac{\partial \left(\frac{1}{4\pi r} \right)}{-\partial r} = \frac{1}{4\pi r^2}$$

1

$$\begin{aligned} & \iint_{\partial B} u \frac{\partial \Gamma}{\partial n} ds \\ &= \iint_{\partial B} u \frac{1}{4\pi r^2} ds \\ &= \frac{1}{4\pi \epsilon^2} \iint_{\partial B} u ds \\ &= \frac{1}{4\pi \epsilon^2} \cdot u(\bar{x}, \bar{y}, \bar{z}) \cdot 4\pi \epsilon^2 \\ &= u(\bar{x}, \bar{y}, \bar{z}) \end{aligned}$$

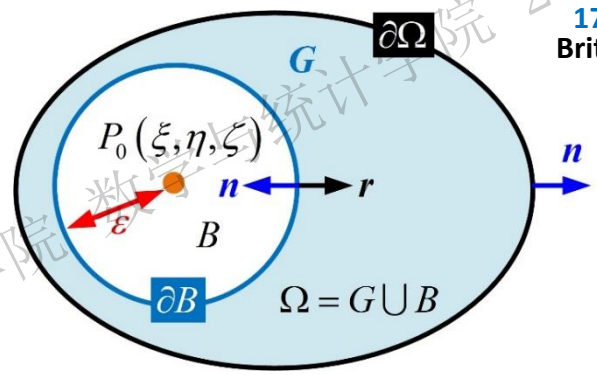
边界 ∂B 上 $r = \epsilon$

中值定理

2

$$\begin{aligned} & \iint_{\partial B} \Gamma \frac{\partial u}{\partial n} ds \\ &= \iint_{\partial B} \frac{1}{4\pi r} \frac{\partial u}{\partial n} ds \\ &= \frac{1}{4\pi \epsilon} \cdot \frac{\partial u}{\partial n}(\bar{x}_1, \bar{y}_1, \bar{z}_1) \cdot 4\pi \epsilon^2 \\ &= \epsilon \frac{\partial u}{\partial n}(\bar{x}_1, \bar{y}_1, \bar{z}_1) \end{aligned}$$

中值定理



$$-\iiint_G \Gamma \Delta u dV = \iint_{\partial\Omega} \left(u \frac{\partial \Gamma}{\partial n} - \Gamma \frac{\partial u}{\partial n} \right) ds + u(\bar{x}, \bar{y}, \bar{z}) - \epsilon \frac{\partial u}{\partial n}(\bar{x}_1, \bar{y}_1, \bar{z}_1)$$

5.1 Green公式

Green公式 G3公式

Green第1公式

$$\iiint_{\Omega} u \Delta v dV = \iint_{\partial \Omega} u \frac{\partial v}{\partial n} ds - \iiint_{\Omega} \nabla u \cdot \nabla v dV$$

Green第2公式

$$\iiint_{\Omega} (u \Delta v - v \Delta u) dV = \iint_{\partial \Omega} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds$$



G.Green
1793-1841
British Math.

当 $\varepsilon \rightarrow 0$ 时, 有 $u(\bar{x}, \bar{y}, \bar{z}) \rightarrow u(\xi, \eta, \zeta)$, $\varepsilon \frac{\partial u}{\partial n}(\bar{x}_1, \bar{y}_1, \bar{z}_1) \rightarrow 0$, $G \rightarrow \Omega \setminus P_0$

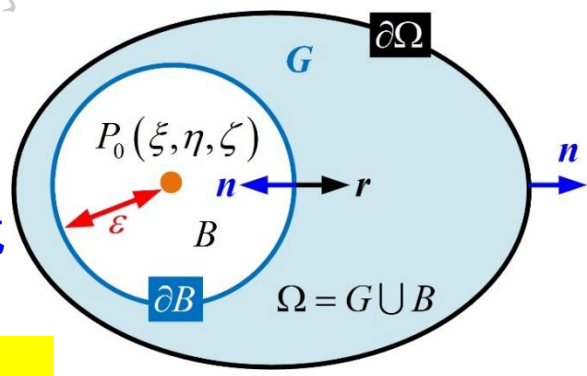
代入有

$$-\iiint_G \Gamma \Delta u dV = \iint_{\partial \Omega} \left(u \frac{\partial \Gamma}{\partial n} - \Gamma \frac{\partial u}{\partial n} \right) ds + u(\bar{x}, \bar{y}, \bar{z}) - \varepsilon \frac{\partial u}{\partial n}(\bar{x}_1, \bar{y}_1, \bar{z}_1)$$
$$-\iiint_{\Omega} \Gamma \Delta u dV = \iint_{\partial \Omega} \left(u \frac{\partial \Gamma}{\partial n} - \Gamma \frac{\partial u}{\partial n} \right) ds + u(\xi, \eta, \zeta)$$

移项得

$$u(\xi, \eta, \zeta) = \iint_{\partial \Omega} \left(\Gamma \frac{\partial u}{\partial n} - u \frac{\partial \Gamma}{\partial n} \right) ds - \iiint_{\Omega} \Gamma \Delta u dV$$

Green第3公式



- 1. 利用 $\Gamma(x, y, z)$ 的奇点得到了 Δu , u , $\frac{\partial u}{\partial n}$ 之间的固有关系
- 2. 说明 $u(\xi, \eta, \zeta)$ 在区域 Ω 内可用 Δu 及边界上的 u , $\frac{\partial u}{\partial n}$ 表示

下面利用Green第3公式 $\rightarrow$ Poisson方程Dirichlet问题 $\rightarrow$ 关键是引入边界信息

思路 G3公式 $\rightarrow$ 基本解/关联边界条件 $\rightarrow$ 引入Green函数 $\rightarrow$ Poisson方程积分形式解

数学物理方程

1. 数学建模和基本原理介绍

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5.1 Green公式

5.2 Laplace方程基本解

5.3 Green函数

5.4 半空间/圆域上的Dirichlet问题

5.2 Laplace方程基本解

预备知识

基本解

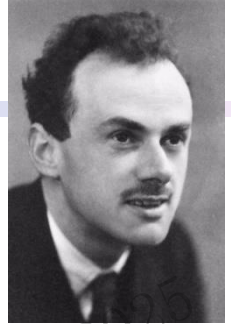
- 1. 基本解 Laplace方程基本解的简称
- 2. 满足Laplace方程/可作为基本解的函数有很多→Green函数理论→类比点电荷产生电势建立基本解→赋予基本解物理意义→为从物理意义出发构造Green函数提供思路

$$\Gamma(P, P_0) = \frac{1}{4\pi} \cdot \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}}$$

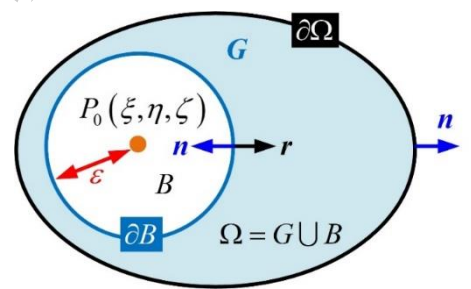
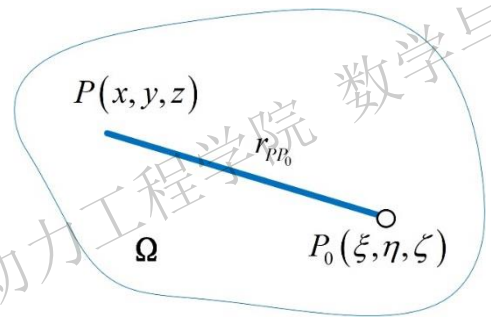
$$\Gamma(P, P_0) = \frac{1}{2\pi} \ln \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2}}$$

Dirac函数

$$\delta(x) = \begin{cases} \infty, & x = 0 \\ 0, & x \neq 0 \end{cases}$$
$$\int_{-\infty}^{+\infty} \delta(x) dx = 1$$



P. Dirac
1902-1984
British Math.
Physical Scientist
1933 Nobel Winner



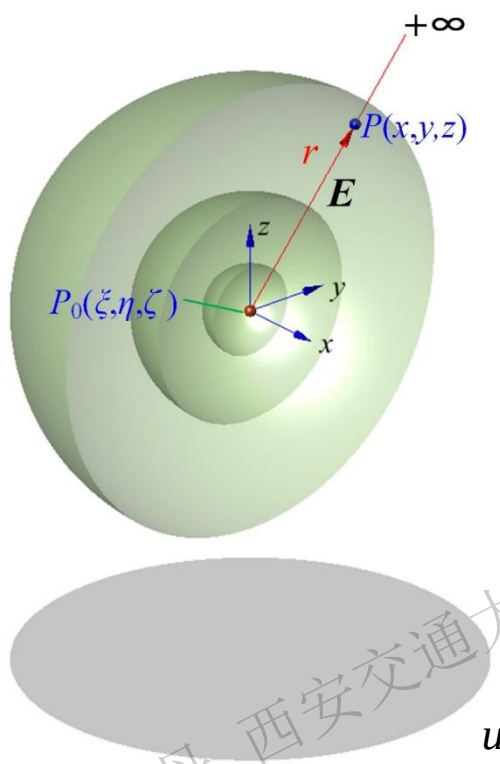
基于距离函数的基本解存在奇点 ↓

$$\Delta \Gamma(P, P_0) = \delta(P, P_0) = \begin{cases} 0, & P \neq P_0 \\ \infty, & P = P_0 \end{cases}$$

- 1. Dirac函数表征了物理量的位置对其空间分布的最大影响
- 2. 本课程引入Dirac函数仅仅为了简化 $\Delta \Gamma$ 表达
- 3. 正是利用奇点→推出Green第3公式→后续计算都在奇点外进行→无需关注奇点

5.2 Laplace方程基本解

预备知识 点电荷电势 三维



Coulomb定律→电荷间相互作用力

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q \cdot q_0}{r^2} \mathbf{r}^0, \quad \epsilon_0 = 8.85 \times 10^{-12} \frac{F}{m}$$

真空介电常数



C.Coulomb
1736-1806
French Phys.

作用力通过电场传递→电场强度

$$E = \frac{F}{q_0} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \mathbf{r}^0$$

- 1. 电场为保守场→可定义电势
- 2. 点电荷 P_0 在点 P 处的电势=单位正电荷从点 P 移动到0电势点的过程中静电力所作的功
- 3. 点电荷产生的电势以 $r=\infty$ 处为0电势参考点

$$u(P, P_0) = \int_{r_{0P}}^{\infty} E \cdot dr = \int_{r_{0P}}^{\infty} \frac{q}{4\pi\epsilon_0 r^2} \cdot dr = \frac{q}{4\pi\epsilon_0} \cdot \left(-\frac{1}{r} \right) \Big|_{r_{0P}}^{\infty} = \frac{q}{\epsilon_0} \cdot \left[\frac{1}{4\pi r_{0P}} \right]$$

当 $P \neq P_0$ 时→ $\Gamma(P, P_0)$ 满足三维Laplace方程→称 $\Gamma(P, P_0)$ 为Laplace方程三维基本解

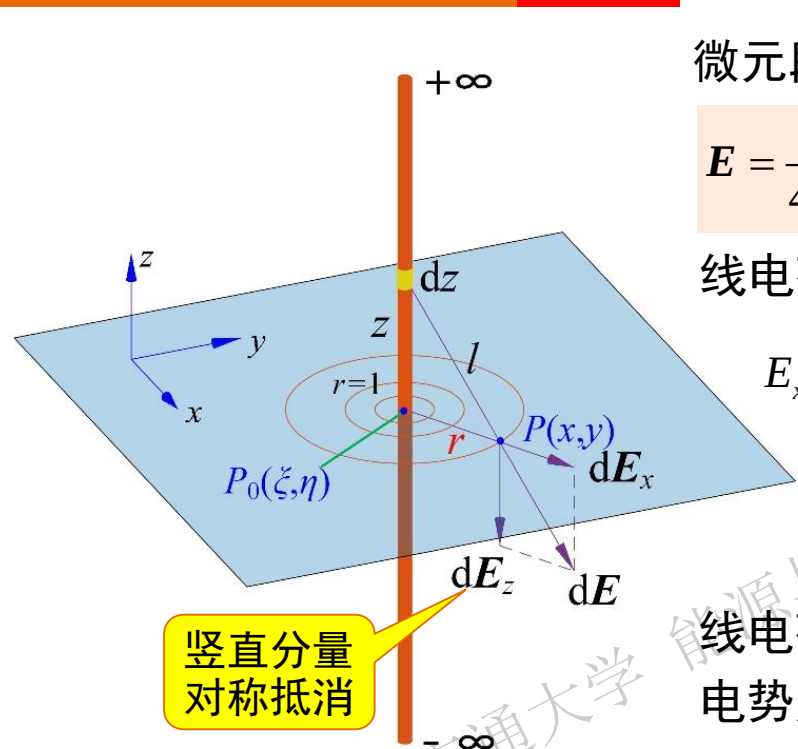
← $\Delta \Gamma = \Gamma_{xx} + \Gamma_{yy} + \Gamma_{zz} = 0$ →

↓
 $\Gamma(P, P_0) = \frac{1}{4\pi r_{0P}}$

5.2 Laplace方程基本解

预备知识 线电荷电势 二维

无限长线电荷电势仅与径向距离 r 有关→二维电势分布



微元段电荷在点 P 处电场水平分量

$$E = \frac{q}{4\pi\epsilon_0 r^2} \mathbf{r}^0 \quad \rightarrow \quad dE_x = \frac{\lambda \cdot dz}{4\pi\epsilon_0 l^2} \cdot \frac{r}{l} = \frac{\lambda}{4\pi\epsilon_0} \cdot \frac{r}{(z^2 + r^2)^{3/2}} \cdot dz$$

电荷线密度

线电荷径向点 P 处总电场水平分量

$$E_x = 2 \cdot \int_0^{+\infty} dE_x = 2 \cdot \int_0^{+\infty} \frac{\lambda}{4\pi\epsilon_0} \cdot \frac{r}{(z^2 + r^2)^{3/2}} \cdot dz$$
$$= \frac{\lambda}{2\pi\epsilon_0} \cdot r \cdot \frac{z}{r^2 \sqrt{z^2 + r^2}} \Big|_0^{+\infty} = \frac{\lambda}{2\pi\epsilon_0 \cdot r}$$

线电荷在点 P 处的电势=单位正电荷从点 P 移动到0电势点的过程中静电力所作的功→以 $r=1$ 处为0电势

$$u(P, P_0) = \int_{r_{P_0P}}^{r=1} E_x \cdot dr = \int_{r_{P_0P}}^{r=1} \frac{\lambda}{2\pi\epsilon_0 \cdot r} \cdot dr = \frac{\lambda}{2\pi\epsilon_0} \ln r \Big|_{r_{P_0P}}^{r=1} = -\frac{\lambda}{2\pi\epsilon_0} \ln r_{P_0P} = \frac{\lambda}{\epsilon_0} \cdot \left[\frac{1}{2\pi} \ln \frac{1}{r_{P_0P}} \right]$$

当 $P \neq P_0$ 时→ $\Gamma(P, P_0)$ 满足二维Laplace方程→称 $\Gamma(P, P_0)$ 为Laplace方程二维基本解

$$\Delta \Gamma = \Gamma_{xx} + \Gamma_{yy} = 0 \quad \leftarrow \quad \Gamma(P, P_0) = \frac{1}{2\pi} \ln \frac{1}{r_{P_0P}}$$

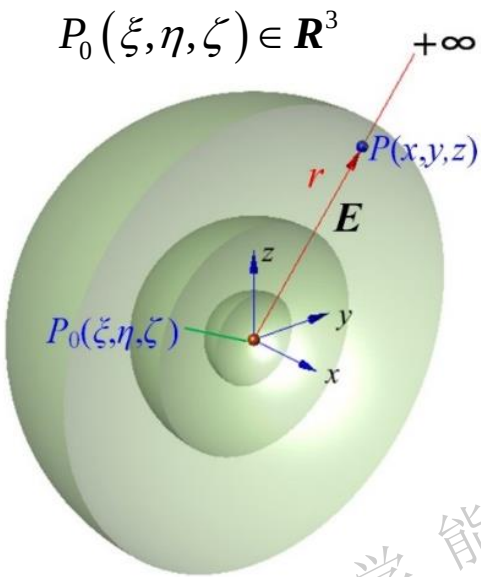
5.2 Laplace方程基本解

基本解

基本解是Laplace方程的基本解

$$\Delta \Gamma(P, P_0) = \delta(P, P_0) = \begin{cases} 0, & P \neq P_0 \\ \infty, & P = P_0 \end{cases}$$

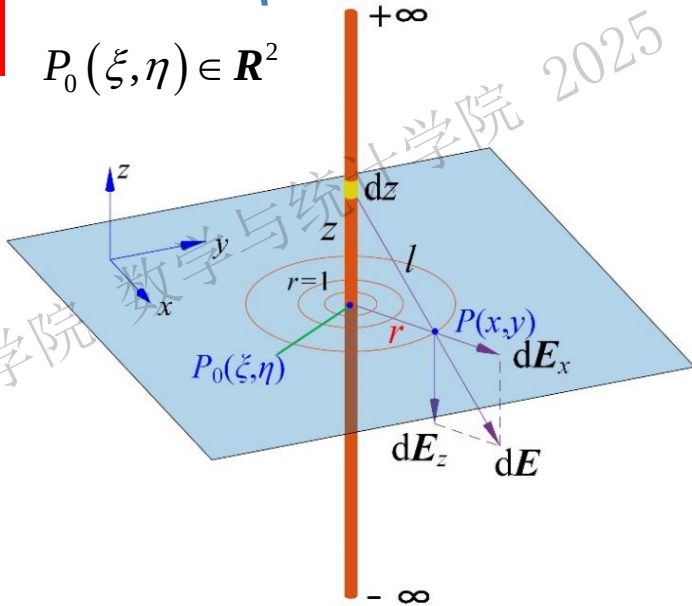
三维



$$\Gamma(P, P_0) = \frac{1}{4\pi} \cdot \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}}$$

三维基本解代表点电荷空间电势

二维



$$\Gamma(P, P_0) = \frac{1}{2\pi} \ln \frac{1}{r_{P_0 P}} = \frac{1}{2\pi} \ln \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2}}$$

二维基本解代表无限长线电荷的平面电势

- 1. 三维/二维基本解均基于距离函数→在 $r=0$, 即 $P=P_0$ 处是奇点
- 2. 对三维/二维基本解的运算任然默认对 (x, y, z) 运算

5.2 Laplace方程基本解

例1 验证函数 $u(x,y) = \frac{1}{2\pi} \ln \frac{1}{r}$, $r = \sqrt{(x-x_0)^2 + (y-y_0)^2}$ 在 $\mathbf{R}^2 \setminus \{(x_0,y_0)\}$ 上是方程 $u_{xx} + u_{yy} = 0$ 的古典解

解

$$u_{xx} = \frac{(x-x_0)^2}{\pi r^4} - \frac{1}{2\pi r^2}, \quad u_{yy} = \frac{(y-y_0)^2}{\pi r^4} - \frac{1}{2\pi r^2} \quad \text{代入原方程得证}$$
$$u_{xx} + u_{yy} = \frac{(x-x_0)^2}{\pi r^4} - \frac{1}{2\pi r^2} + \frac{(y-y_0)^2}{\pi r^4} - \frac{1}{2\pi r^2} = \frac{r^2}{\pi r^4} - \frac{1}{\pi r^2} = 0$$

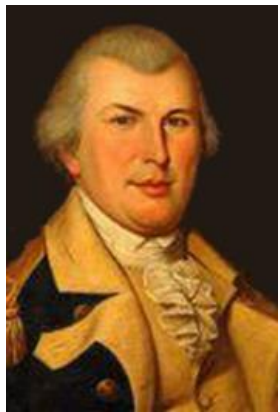
Laplace方程在圆域/球域上的基本解

例2 验证函数 $u(x,y,z) = \frac{1}{4\pi r}$, $r = \sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}$ 在 $\mathbf{R}^3 \setminus \{(x_0,y_0,z_0)\}$ 上是方程 $u_{xx} + u_{yy} + u_{zz} = 0$ 的古典解

解

$$\left. \begin{aligned} u_{xx} &= \frac{2(x-x_0)^2 - (y-y_0)^2 - (z-z_0)^2}{4\pi r^5} \\ u_{yy} &= \frac{2(y-y_0)^2 - (x-x_0)^2 - (z-z_0)^2}{4\pi r^5} \\ u_{zz} &= \frac{2(z-z_0)^2 - (x-x_0)^2 - (y-y_0)^2}{4\pi r^5} \end{aligned} \right\} u_{xx} + u_{yy} + u_{zz} = 0$$

G.Green
1793-1841
British Math.
Physical Scientist



1. 数学建模和基本原理介绍

2. 分离变量法

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4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

5.1 Green公式

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5.3 Green函数

5.4 半空间/圆域上的Dirichlet问题

5.3 Green函数

Green函数法解Poisson方程

Poisson方程定解问题
$$\begin{cases} -\Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ u(x, y, z) = \varphi(x, y, z) & (x, y, z) \in \partial\Omega \end{cases}$$

本质：借助定解条件→化简Green第3公式

设存在解 $u(x, y, z) \in C^2(\Omega) \cap C^1(\bar{\Omega})$, $P_0(\xi, \eta, \zeta) \in \mathbf{R}^3$

根据Green第3公式

$$u(\xi, \eta, \zeta) = \iint_{\partial\Omega} \left(\Gamma \frac{\partial u}{\partial n} - u \frac{\partial \Gamma}{\partial n} \right) ds - \iiint_{\Omega} \Gamma \Delta u dV$$

$$u(\xi, \eta, \zeta) = \iint_{\partial\Omega} \Gamma \frac{\partial u}{\partial n} ds - \iint_{\partial\Omega} u \frac{\partial \Gamma}{\partial n} ds - \iiint_{\Omega} \Gamma \Delta u dV$$

边界条件

自由项

$$u(\xi, \eta, \zeta) = \iint_{\partial\Omega} \Gamma \frac{\partial u}{\partial n} ds - \iint_{\partial\Omega} \varphi \frac{\partial \Gamma}{\partial n} ds - \iiint_{\Omega} \Gamma (-f) dV$$

计算非常困难
→基本解 Γ 下手→叠加辅助函数 h →使得在边界上 $\Gamma + h = 0$

5.3 Green函数

Green函数法解Poisson方程

Poisson方程
定解问题

$$\begin{cases} -\Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ u(x, y, z) = \varphi(x, y, z) & (x, y, z) \in \partial\Omega \end{cases}$$

$$\begin{aligned} \iiint_{\Omega} u \Delta v dV &= \iint_{\partial\Omega} u \frac{\partial v}{\partial n} ds - \iiint_{\Omega} \nabla u \cdot \nabla v dV \\ \iiint_{\Omega} (u \Delta v - v \Delta u) dV &= \iint_{\partial\Omega} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds \\ u &= \iint_{\partial\Omega} \left(\Gamma \frac{\partial u}{\partial n} - u \frac{\partial \Gamma}{\partial n} \right) ds - \iiint_{\Omega} \Gamma \Delta u dV \end{aligned}$$

$$u(\xi, \eta, \zeta) = \iint_{\partial\Omega} \Gamma \frac{\partial u}{\partial n} ds - \iint_{\partial\Omega} u \frac{\partial \Gamma}{\partial n} ds - \iiint_{\Omega} \Gamma \Delta u dV \quad \text{①}$$

① + ② 有

$$u(\xi, \eta, \zeta) = \iint_{\partial\Omega} \left[(\Gamma + h) \frac{\partial u}{\partial n} - u \frac{\partial (\Gamma + h)}{\partial n} \right] ds - \iiint_{\Omega} (\Gamma + h) \Delta u dV$$

设 $h(x, y, z)$ 是如下定解问题的解

$$\begin{cases} -\Delta h = 0, & (x, y, z) \in \Omega \\ h(x, y, z) = -\Gamma(x, y, z) & (x, y, z) \in \partial\Omega \end{cases}$$

Green第2公式中取 $v = h(x, y, z)$

$$\begin{aligned} \iiint_{\Omega} (u \Delta h - h \Delta u) dV &= \iint_{\partial\Omega} \left(u \frac{\partial h}{\partial n} - h \frac{\partial u}{\partial n} \right) ds \\ \text{=0} \quad - \iiint_{\Omega} h \Delta u dV &= - \iint_{\partial\Omega} h \frac{\partial u}{\partial n} ds + \iint_{\partial\Omega} u \frac{\partial h}{\partial n} ds \end{aligned}$$

得 $0 = \iint_{\partial\Omega} h \frac{\partial u}{\partial n} ds - \iint_{\partial\Omega} u \frac{\partial h}{\partial n} ds - \iiint_{\Omega} h \Delta u dV \quad \text{②}$

Green函数

$$G(x, y, z) = \Gamma(x, y, z) + h(x, y, z)$$

边界上=0

$$\begin{aligned} u(\xi, \eta, \zeta) &= \iint_{\partial\Omega} \left[G \frac{\partial u}{\partial n} - u \frac{\partial G}{\partial n} \right] ds - \iiint_{\Omega} G \Delta u dV \\ &= - \iint_{\partial\Omega} u \frac{\partial G}{\partial n} ds - \iiint_{\Omega} G \Delta u dV \end{aligned}$$

Poisson方程第一类边界下的解可表示为

$$u(\xi, \eta, \zeta) = - \iint_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV$$

5.3 Green函数

Green函数 Poisson方程定解问题

$$\begin{cases} -\Delta u = f(x,y,z), & (x,y,z) \in \Omega \\ u(x,y,z) = \varphi(x,y,z) & (x,y,z) \in \partial\Omega \end{cases}$$

基本解→G3

$$u(\xi,\eta,\zeta) = -\iint_{\partial\Omega} \varphi \frac{\partial \Gamma}{\partial n} ds - \iiint_{\Omega} \Gamma \Delta u dV + \iint_{\partial\Omega} \Gamma \frac{\partial u}{\partial n} ds$$

边界上=0

$$u(\xi,\eta,\zeta) = -\iint_{\partial\Omega} u \frac{\partial G}{\partial n} ds - \iiint_{\Omega} G \Delta u dV + \iint_{\partial\Omega} G \frac{\partial u}{\partial n} ds$$

$$u(\xi,\eta,\zeta) = -\iint_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV$$

庆幸 特殊区域→球形/圆域/半空间
→基本解物理意义线性组合→构造
Green函数→把原Poisson方程的解用
积分表达出来

思路 借助辅助定解问题→引入Green函数→
消去G3边界导数项→得原Poisson定解问题解

$$\begin{cases} -\Delta h = 0, & (x,y,z) \in \Omega \\ h(x,y,z) = -\Gamma(x,y,z) & (x,y,z) \in \partial\Omega \end{cases}$$

辅助定解问题

$$G(x,y,z) = \Gamma(x,y,z) + h(x,y,z)$$

Green函数G(x,y,z) 是定解问题的解

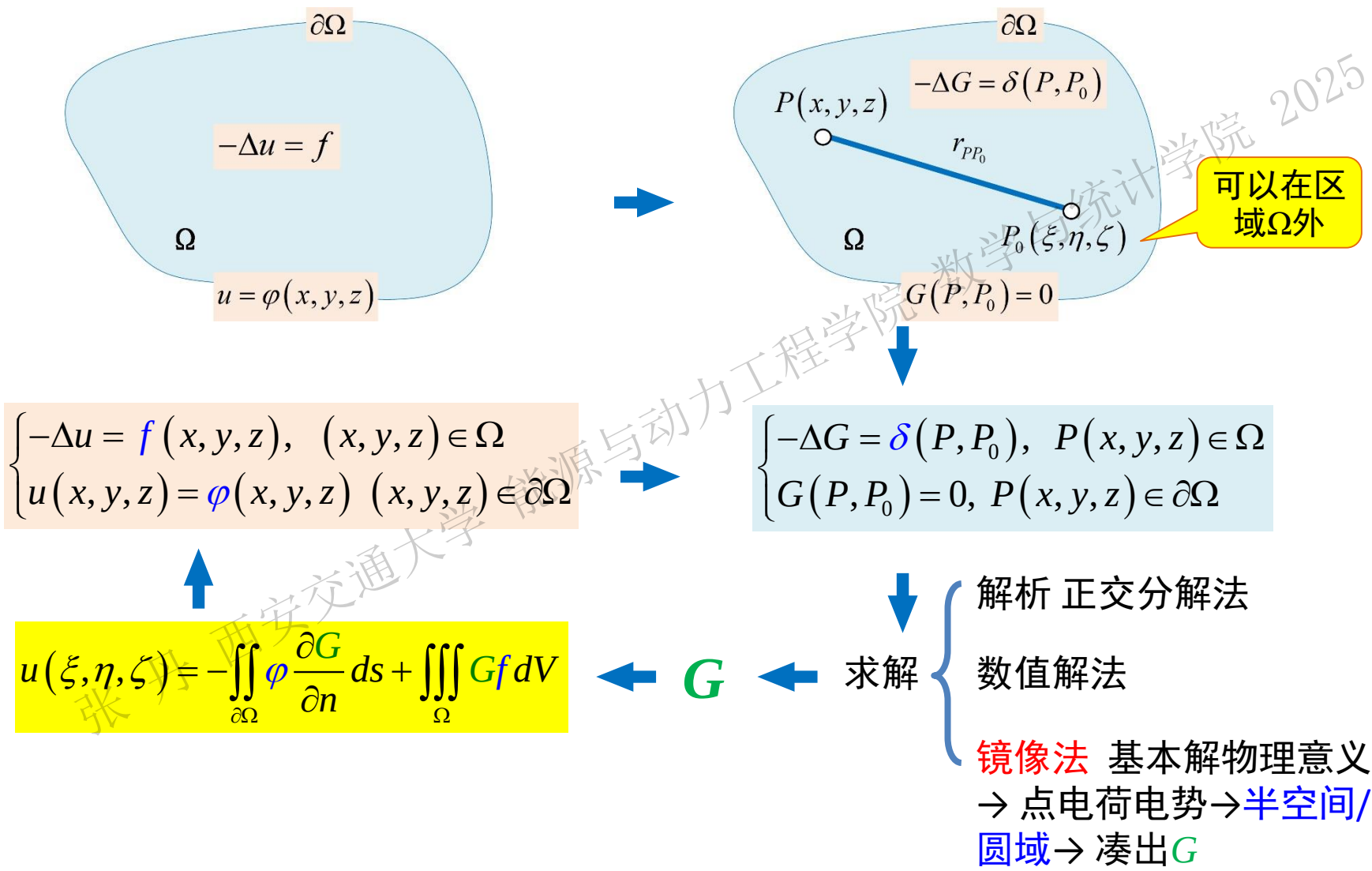
$$\begin{cases} -\Delta G = -(\Delta \Gamma + \Delta h) = \delta(x,y,z), & (x,y,z) \in \Omega \\ G(x,y,z) = \Gamma(x,y,z) + h(x,y,z) = 0, & (x,y,z) \in \partial\Omega \end{cases}$$

高明 Green函数仅依赖于区域Ω →与原
Poisson方程边界/自由项无关→若求得G
→可一劳永逸求解任何边界下Poisson方程

遗憾 任意边界下Green函数定解问题求解
很困难

5.3 Green函数

Green函数法求解步骤



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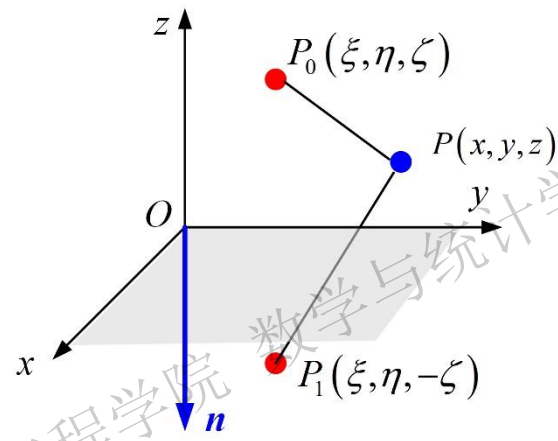
5.4 半空间/圆域上的Dirichlet问题

例1 半空间Dirichlet问题 设有半空间 $\Omega = \{(x, y, z) \mid z > 0\}$, $\partial\Omega = \{(x, y, z) \mid z = 0\}$

$$\begin{cases} -\Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ u(x, y, 0) = \varphi(x, y) & (x, y) \in \partial\Omega \end{cases}$$

解 借助Green函数, 定解问题的通解为

$$u(\xi, \eta, \zeta) = - \iint_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV$$



ξ /'ksai/
 ζ /'zi:tə/

关键问题 在题设区域下构造 $G(x, y, z)$ 使其满足

$$\begin{cases} -\Delta G = \delta(P, P_0), & P(x, y, z) \in \Omega \\ G(P, P_0) = 0, & P(x, y, z) \in \partial\Omega \end{cases}$$

关键是满足
边界条件

基本解 $\Gamma(P, P_0) = \frac{1}{4\pi r_{P_0P}}$ 满足Laplace方程 $-\Delta \Gamma(P, P_0) = \delta(P, P_0)$, $P(x, y, z) \in \Omega$

基本解**满足方程/不满足边界条件**→但基本解线性组合依然满足方程→根据物理意义线性组合多个基本解→实现边界条件→**Green函数**→代入积分形式解

5.4 半空间/圆域上的Dirichlet问题

例1 半空间Dirichlet问题

设有半空间 $\Omega = \{(x, y, z) \mid z > 0\}$, $\partial\Omega = \{(x, y, z) \mid z = 0\}$

$$\begin{cases} -\Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ u(x, y, 0) = \varphi(x, y) & (x, y) \in \partial\Omega \end{cases}$$

基本解 $\Gamma(P, P_0) = \frac{1}{4\pi r_{P_0P}}$

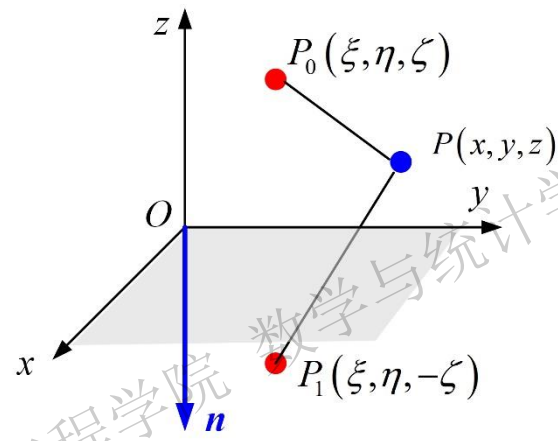
相当于在 $P_0(\xi, \eta, \zeta)$ 放置正电荷产生的电势

若在对称点 $P_1(\xi, \eta, -\zeta)$ 放置等电量正电荷 → 对称性可保证边界 $z=0$ 平面电势为0

点电荷 $P_1(\xi, \eta, -\zeta)$ 产生的电位分布满足 $-\Delta \Gamma(P, P_1) = 0, P \in \Omega$

可取Green函数为两点电位之差 $G(P, P_0) = \Gamma(P, P_0) - \Gamma(P, P_1) = \delta(P, P_0), P(x, y, z) \in \Omega$

表示 P_1, P_0 电荷产生电位在 $P(x, y, z)$ 处叠加



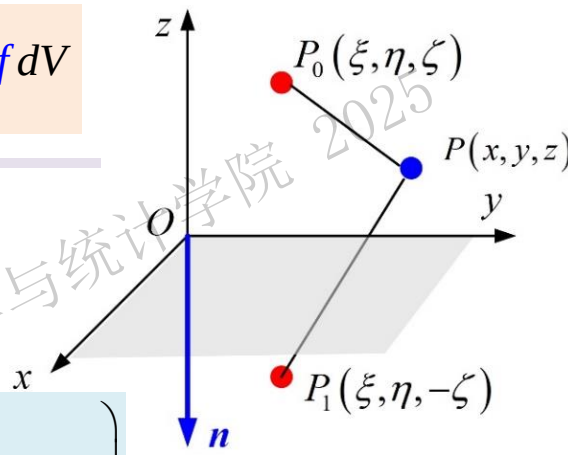
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例1 半空间Dirichlet问题

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$$\begin{cases} -\Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ u(x, y, 0) = \varphi(x, y) & (x, y) \in \partial\Omega \end{cases}$$

$$u(\xi, \eta, \zeta) = -\iint_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV$$



取Green函数为

$$G(P, P_0) = \Gamma(P, P_0) - \Gamma(P, P_1) = \frac{1}{4\pi} \left(\frac{1}{r_{PP_0}} - \frac{1}{r_{PP_1}} \right)$$

$$G(x, y, z) = \frac{1}{4\pi} \left(\frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}} - \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + [z-(-\zeta)]^2}} \right)$$

半空间xOy平面外法线 $n = -z$

$$\begin{aligned} \frac{\partial G}{\partial n} \Big|_{\partial\Omega} &= - \frac{\partial G}{\partial z} \Big|_{z=0} = - \frac{1}{4\pi} \left\{ \frac{-\frac{1}{2} \cdot 2(z-\zeta)}{\left[(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2 \right]^{\frac{3}{2}}} - \frac{-\frac{1}{2} \cdot 2(z-(-\zeta))}{\left[(x-\xi)^2 + (y-\eta)^2 + (z-(-\zeta))^2 \right]^{\frac{3}{2}}} \right\}_{z=0} \\ &= - \frac{1}{4\pi} \left\{ \frac{\zeta}{\left[(x-\xi)^2 + (y-\eta)^2 + \zeta^2 \right]^{\frac{3}{2}}} - \frac{-\zeta}{\left[(x-\xi)^2 + (y-\eta)^2 + \zeta^2 \right]^{\frac{3}{2}}} \right\} = - \frac{1}{2\pi} \cdot \frac{\zeta}{\left[(x-\xi)^2 + (y-\eta)^2 + \zeta^2 \right]^{\frac{3}{2}}} \end{aligned}$$

5.4 半空间/圆域上的Dirichlet问题

例1 半空间Dirichlet问题

设有半空间 $\Omega = \{(x, y, z) \mid z > 0\}$, $\partial\Omega = \{(x, y, z) \mid z = 0\}$

$$\begin{cases} -\Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ u(x, y, 0) = \varphi(x, y) & (x, y) \in \partial\Omega \end{cases}$$

$$u(\xi, \eta, \zeta) = -\iint_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV$$

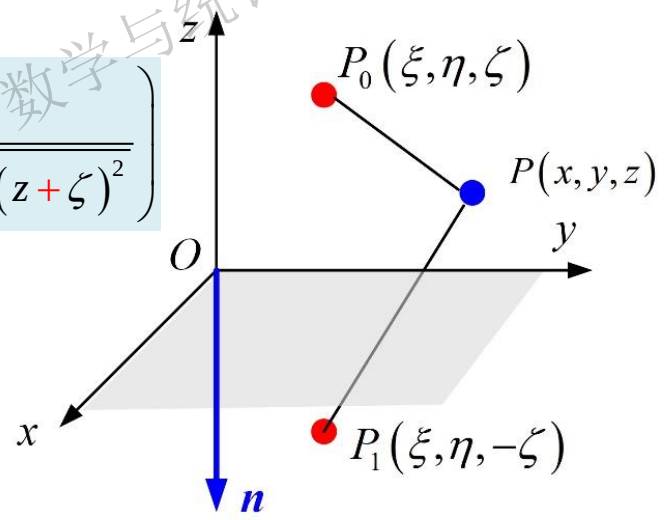
取Green函数为 $G(P, P_0) = \Gamma(P, P_0) - \Gamma(P, P_1) = \frac{1}{4\pi} \left(\frac{1}{r_{PP_0}} - \frac{1}{r_{PP_1}} \right)$

$$G(x, y, z) = \frac{1}{4\pi} \left(\frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}} - \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z+\zeta)^2}} \right)$$

$$\left. \frac{\partial G}{\partial n} \right|_{\partial\Omega} = - \left. \frac{\partial G}{\partial z} \right|_{z=0} = - \frac{1}{2\pi} \frac{\zeta}{\left[(x-\xi)^2 + (y-\eta)^2 + \zeta^2 \right]^{\frac{3}{2}}}$$

代入积分形式解

$$\begin{aligned} u(\xi, \eta, \zeta) &= -\iint_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f \cdot dV \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{\varphi(x, y) \cdot \zeta}{\left[(x-\xi)^2 + (y-\eta)^2 + \zeta^2 \right]^{\frac{3}{2}}} dx dy + \int_0^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} G(x, y, z) \cdot f(x, y, z) dx dy dz \end{aligned}$$



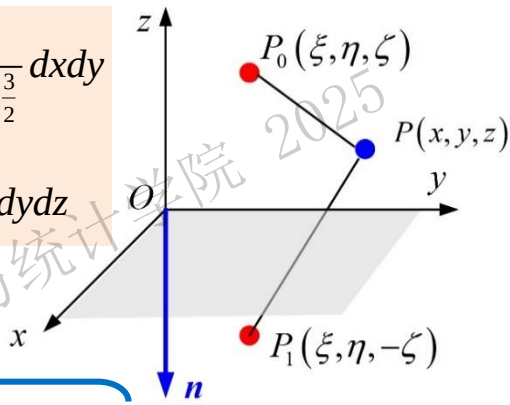
5.3 半空间/圆域上的Dirichlet问题

例1 半空间Dirichlet问题

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$$\begin{cases} -\Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ u(x, y, 0) = \varphi(x, y) & (x, y) \in \partial\Omega \end{cases}$$

$$u(\xi, \eta, \zeta) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{\varphi(x, y) \cdot \zeta}{\left[(x-\xi)^2 + (y-\eta)^2 + \zeta^2 \right]^{\frac{3}{2}}} dx dy + \int_0^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} G(x, y, z) \cdot f(x, y, z) dx dy dz$$



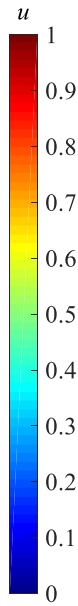
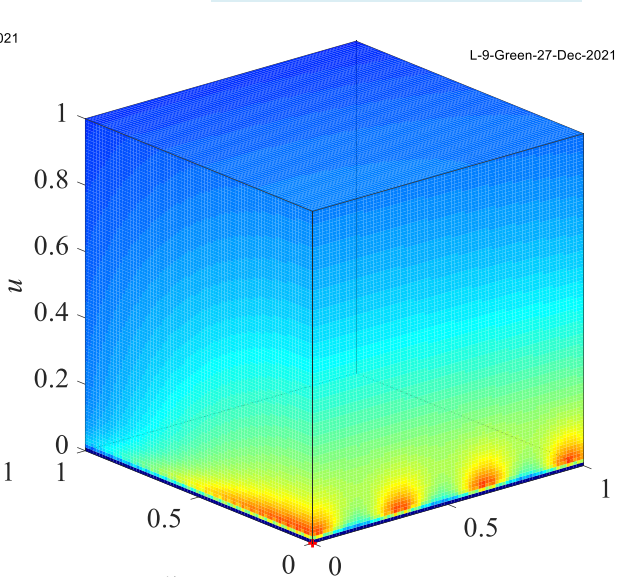
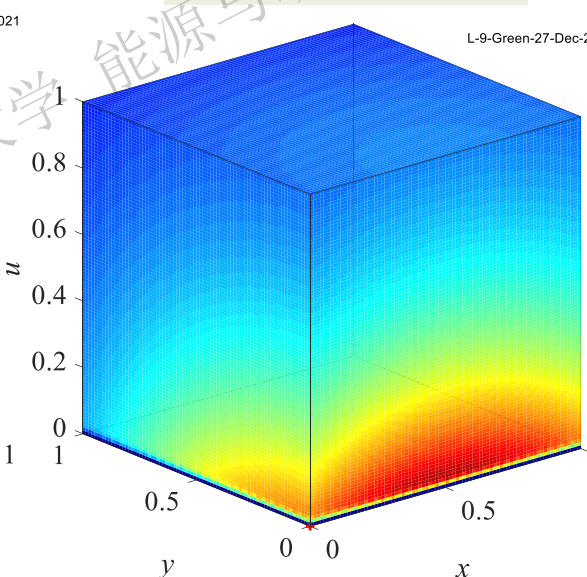
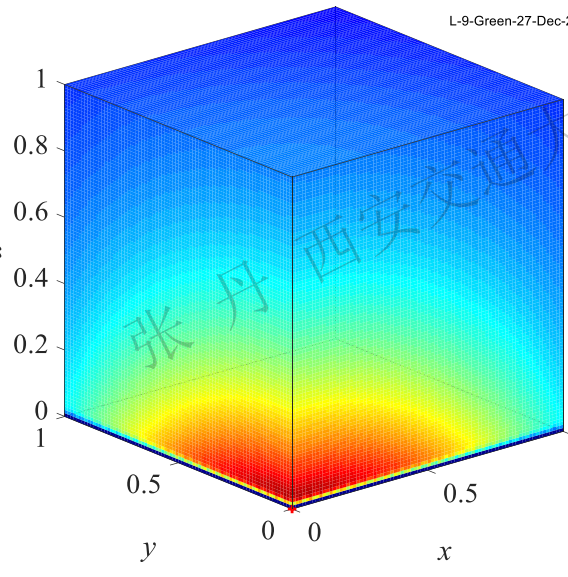
讨论

$$G(x, y, z) = \frac{1}{4\pi} \left(\frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}} - \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2 + (z+\zeta)^2}} \right)$$

$$\begin{aligned} \varphi(x, y) &= e^{-(x^2+y^2)} \\ f(x, y, z) &= 0 \end{aligned}$$

$$\begin{aligned} \varphi(x, y) &= e^{-[(x-0.5)^2+y^2]} \\ f(x, y, z) &= 0 \end{aligned}$$

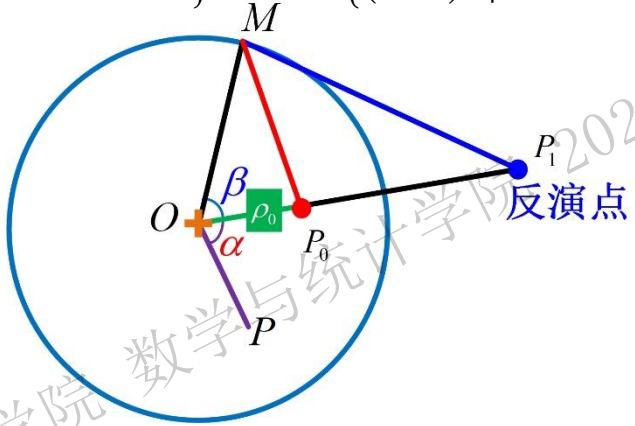
$$\begin{aligned} \varphi(x, y) &= e^{-[\sin^2(10x)+y^2]} \\ f(x, y, z) &= 0 \end{aligned}$$



5.4 半空间/圆域上的Dirichlet问题

例2 圆域Dirichlet问题 设有圆域 $\Omega = \{(x,y) \mid x^2 + y^2 < R^2\}$, $\partial\Omega = \{(x,y) \mid x^2 + y^2 = R^2\}$

$$\begin{cases} -\Delta u = f(x,y), & (x,y) \in \Omega \\ u(x,y) = g(x,y) & (x,y) \in \partial\Omega \end{cases}$$



解 第1步 写出积分形式解

$$u(\xi, \eta, \zeta) = -\iint_{\partial\Omega} \phi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV$$

按二维化简
两个积分

$$u(\xi, \eta) = -\int_{\partial\Omega} \phi \frac{\partial G}{\partial n} ds + \iint_{\Omega} G f \cdot dV$$

二维

关键问题 在题设区域下构造 $G(x,y)$

$$\begin{cases} -\Delta G = \delta(P, P_0), & P(x,y) \in \Omega & \textcircled{1} \\ G(P, P_0) = 0, & P(x,y) \in \partial\Omega & \textcircled{2} \end{cases}$$

二维**基本解**为

$$\Gamma(P, P_0) = \frac{1}{2\pi} \ln \frac{1}{r_{P_0P}} = \frac{1}{2\pi} \ln \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2}}$$

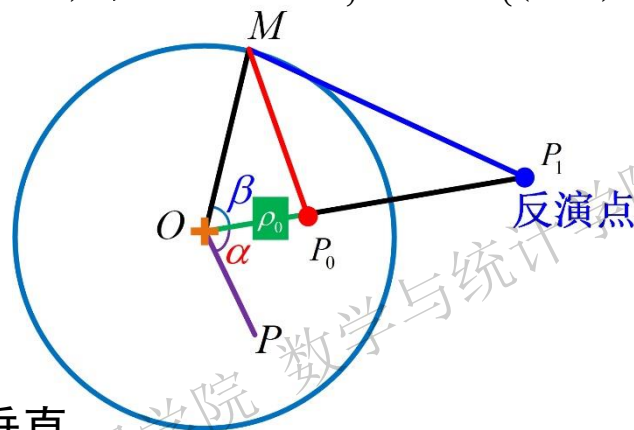
基本解满足方程 ①, 但不满足边界条件 ② → 根据基本解物理意义 → 基本解
线性组合 → 构造满足边界条件的Green函数

5.4 半空间/圆域上的Dirichlet问题

例2 圆域Dirichlet问题

设有圆域 $\Omega = \{(x, y) \mid x^2 + y^2 < R^2\}$, $\partial\Omega = \{(x, y) \mid x^2 + y^2 = R^2\}$

$$\begin{cases} -\Delta u = \textcolor{blue}{f}(x, y), & (x, y) \in \Omega \\ u(x, y) = \textcolor{blue}{g}(x, y) & (x, y) \in \partial\Omega \end{cases}$$



第2步 确定对称电荷位置

基本解 $\Gamma(P, P_0)$ 物理意义表示圆域内 P_0 处垂直放置单位电量的无限长线电荷产生的平面电势

沿径向延长 OP_0 至圆外 P_1

若径向距离满足 $|OP_0| \cdot |OP_1| = R^2$ 称 P_1 是 P_0 关于圆周的反演点

利用反演点可大幅化简距离函数

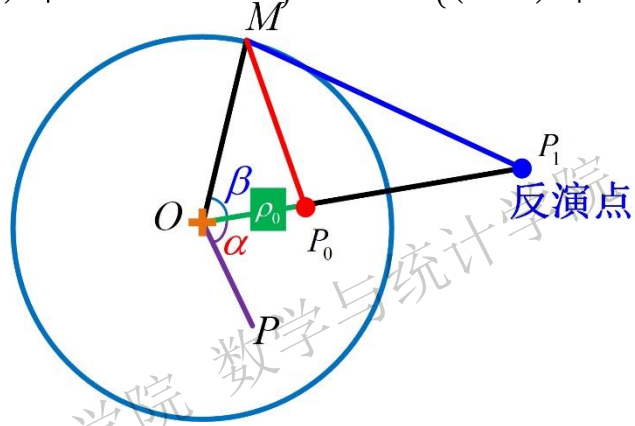
在反演点放置强度为 q^- 的负电荷, 产生电位分布满足 $\Gamma(P, P_1) = -\frac{1}{2\pi} \ln \left(q^- \frac{1}{r_{BP}} \right)$

待定

5.4 半空间/圆域上的Dirichlet问题

例2 圆域Dirichlet问题 设有圆域 $\Omega = \{(x, y) \mid x^2 + y^2 < R^2\}$, $\partial\Omega = \{(x, y) \mid x^2 + y^2 = R^2\}$

$$\begin{cases} -\Delta u = f(x, y), & (x, y) \in \Omega \\ u(x, y) = g(x, y) & (x, y) \in \partial\Omega \end{cases}$$



第3步 确定对称电荷电量

圆周上任取一点 M

由 $|OP_0| \cdot |OP_1| = R^2$ 有 $\frac{|OP_0|}{R} = \frac{R}{|OP_1|}$, 共享角 β $\Rightarrow$ 三角形 $\triangle OP_0M \sim \triangle OP_1M$ 相似

根据相似性 $\frac{|P_0M|}{|OP_0|} = \frac{|P_1M|}{R} \Rightarrow \frac{1}{|P_0M|} = \frac{R}{|OP_0|} \cdot \frac{1}{|P_1M|} \Rightarrow$ 两个电荷到点 M 距离的倒数有确定比例关系

两点电荷在任意 M 点的电势之和为0

$$\Gamma(M, P_0) - \Gamma(M, P_1) = \frac{1}{2\pi} \ln\left(\frac{1}{r_{P_0M}}\right) - \frac{1}{2\pi} \ln\left(\frac{R}{|OP_0|} \cdot \frac{1}{r_{P_1M}}\right) = 0$$

P_0 处为单位正电荷 $\rightarrow$ 只需在反演点放置电量为 $q^- = \frac{R}{|OP_0|}$ 负电荷可使圆周电势为0

5.4 半空间/圆域上的Dirichlet问题

例2 圆域Dirichlet问题 设有圆域 $\Omega = \{(x,y) \mid x^2 + y^2 < R^2\}$, $\partial\Omega = \{(x,y) \mid x^2 + y^2 = R^2\}$

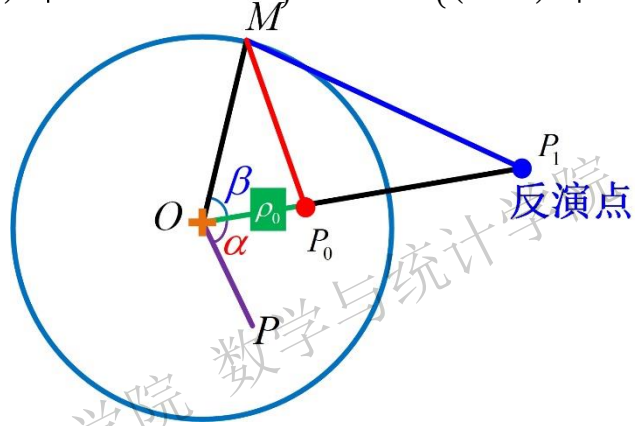
$$\begin{cases} -\Delta u = f(x,y), & (x,y) \in \Omega \\ u(x,y) = g(x,y) & (x,y) \in \partial\Omega \end{cases}$$

第4步 化简Green函数

圆域内任意取点 $P \rightarrow$ 可定义Green函数为

$$\begin{aligned} G(P,P_0) &= \Gamma(P,P_0) - \Gamma(P,P_1) \\ &= \frac{1}{2\pi} \ln\left(\frac{1}{r_{P_0P}}\right) - \frac{1}{2\pi} \ln\left(\frac{R}{|OP_0|} \frac{1}{r_{P_1P}}\right) \\ &= \frac{1}{2\pi} \ln\left(\frac{1}{r_{P_0P}}\right) - \frac{1}{2\pi} \ln\left(\frac{1}{r_{P_1P}}\right) - \frac{1}{2\pi} \ln\left(\frac{R}{|OP_0|}\right) \\ &= \frac{1}{2\pi} \ln\left(\frac{r_{P_1P}}{r_{P_0P}}\right) - \frac{1}{2\pi} \ln\left(\frac{R}{|OP_0|}\right) \end{aligned}$$

直角系内难以继续化简→距离仅与角度有关→极坐标下化简



5.4 半空间/圆域上的Dirichlet问题

例2 圆域Dirichlet问题

设有圆域 $\Omega = \{(x, y) \mid x^2 + y^2 < R^2\}$, $\partial\Omega = \{(x, y) \mid x^2 + y^2 = R^2\}$

$$\begin{cases} -\Delta u = f(x, y), & (x, y) \in \Omega \\ u(x, y) = g(x, y) & (x, y) \in \partial\Omega \end{cases}$$

$$u(\xi, \eta) = -\iint_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV$$

第4步 化简Green函数

极坐标下化简Green函数,记 $P_0(\rho_0, \theta_0)$, $P_1\left(\frac{R^2}{\rho_0}, \theta_0\right)$

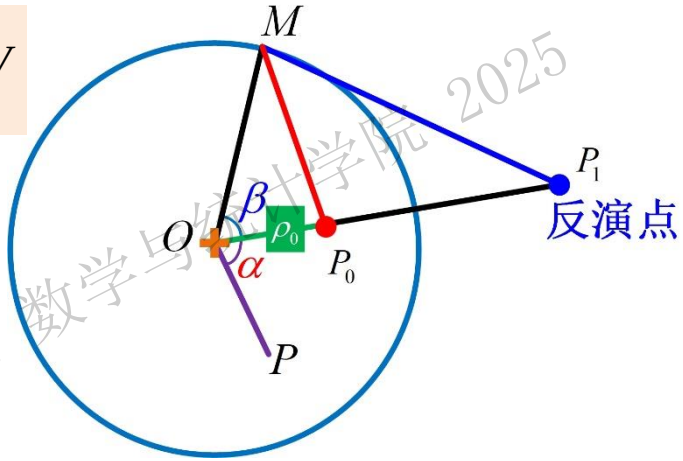
记圆内任意点为 $P(\rho, \theta)$, 设与 P_0 极角差为 α

$$r_{P_0P} = |P, P_0| = \left(\rho_0^2 + \rho^2 - 2\rho_0\rho \cos \alpha\right)^{\frac{1}{2}}$$

$$r_{P_1P} = |P, P_1| = \left[\left(\frac{R^2}{\rho_0}\right)^2 + \rho^2 - 2\frac{R^2}{\rho_0}\rho \cos \alpha\right]^{\frac{1}{2}}$$

$$= \frac{1}{\rho_0} \left[R^4 + \rho_0^2 \rho^2 - 2\rho_0 \rho R^2 \cos \alpha \right]^{\frac{1}{2}}$$

极角差满足 $\alpha = \theta_0 - \theta$



代入Green函数

$$G(P, P_0) = \frac{1}{2\pi} \ln \left(\frac{r_{P_1P}}{r_{P_0P}} \right) - \frac{1}{2\pi} \ln \left(\frac{R}{\rho_0} \right)$$

极坐标下的Green函数

$$G(\rho, \theta) = -\frac{1}{4\pi} \ln \frac{\rho_0^2 R^2 + \rho^2 R^2 - 2\rho_0 \rho R^2 \cos(\theta_0 - \theta)}{R^4 + \rho_0^2 \rho^2 - 2\rho_0 \rho R^2 \cos(\theta_0 - \theta)}$$

圆域边界外法线与径向相同

$$\left. \frac{\partial G}{\partial n} \right|_{\partial\Omega} = \left. \frac{\partial G}{\partial \rho} \right|_{\rho=R} = -\frac{1}{2\pi R} \cdot \frac{R^2 - \rho_0^2}{R^2 + \rho_0^2 - 2\rho_0 R \cos(\theta_0 - \theta)}$$

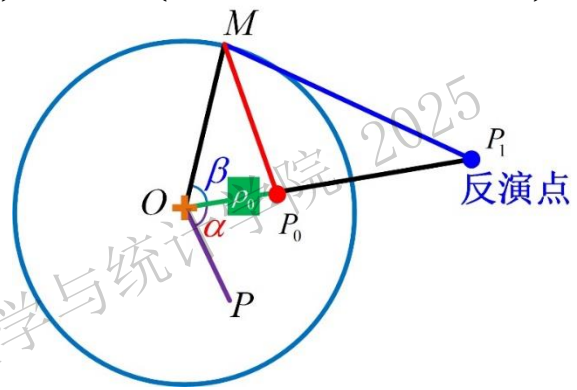
5.4 半空间/圆域上的Dirichlet问题

例2 圆域Dirichlet问题 设有圆域 $\Omega = \{(x, y) \mid x^2 + y^2 < R^2\}$, $\partial\Omega = \{(x, y) \mid x^2 + y^2 = R^2\}$

$$\begin{cases} -\Delta u = f(x, y), & (x, y) \in \Omega \\ u(x, y) = g(x, y) & (x, y) \in \partial\Omega \end{cases}$$

$$G(\rho, \theta) = -\frac{1}{4\pi} \ln \frac{\rho_0^2 R^2 + \rho^2 R^2 - 2\rho_0 \rho R^2 \cos(\theta_0 - \theta)}{R^4 + \rho_0^2 \rho^2 - 2\rho_0 \rho R^2 \cos(\theta_0 - \theta)}$$

$$\left. \frac{\partial G}{\partial n} \right|_{\partial\Omega} = \left. \frac{\partial G}{\partial \rho} \right|_{\rho=R} = -\frac{1}{2\pi R} \cdot \frac{R^2 - \rho_0^2}{R^2 + \rho_0^2 - 2\rho_0 R \cos(\theta_0 - \theta)}$$



$$u(\xi, \eta) = -\iint_{\partial\Omega} \phi \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV$$

圆周边界条件用极坐标表示 $\varphi(\theta) = g(R \cos \theta, R \sin \theta)$

原定解问题的解

$$\begin{aligned} u(\rho_0, \theta_0) &= -\iint_{\partial\Omega} \phi(\theta) \frac{\partial G}{\partial n} ds + \iiint_{\Omega} G f dV \\ &= \frac{1}{2\pi} \cdot \int_0^{2\pi} \frac{(R^2 - \rho_0^2) \phi(\theta)}{R^2 + \rho_0^2 - 2\rho_0 R \cos(\theta_0 - \theta)} d\theta \\ &\quad - \frac{1}{4\pi} \int_0^{2\pi} \int_0^R f(\rho \cos \theta, \rho \sin \theta) \ln \frac{\rho_0^2 R^2 + \rho^2 R^2 - 2\rho_0 \rho R^2 \cos(\theta_0 - \theta)}{R^4 + \rho_0^2 \rho^2 - 2\rho_0 \rho R^2 \cos(\theta_0 - \theta)} \rho d\rho d\theta \end{aligned}$$

5.4 半空间/圆域上的Dirichlet问题

例2 圆域Dirichlet问题

设有圆域 $\Omega = \{(x, y) \mid x^2 + y^2 < R^2\}$, $\partial\Omega = \{(x, y) \mid x^2 + y^2 = R^2\}$

$$\begin{cases} -\Delta u = f(x, y), & (x, y) \in \Omega \\ u(x, y) = g(x, y) & (x, y) \in \partial\Omega \end{cases}$$

讨论

$$u(\rho_0, \theta_0) = \frac{1}{2\pi} \cdot \int_0^{2\pi} \frac{(R^2 - \rho_0^2)\varphi(\theta)}{R^2 + \rho_0^2 - 2\rho_0 R \cos(\theta_0 - \theta)} d\theta$$
$$-\frac{1}{4\pi} \int_0^{2\pi} \int_0^R f(\rho \cos \theta, \rho \sin \theta) \ln \frac{\rho_0^2 R^2 + \rho^2 R^2 - 2\rho_0 \rho R^2 \cos(\theta_0 - \theta)}{R^4 + \rho_0^2 \rho^2 + 2\rho_0 \rho R^2 \cos(\theta_0 - \theta)} \rho d\rho d\theta$$

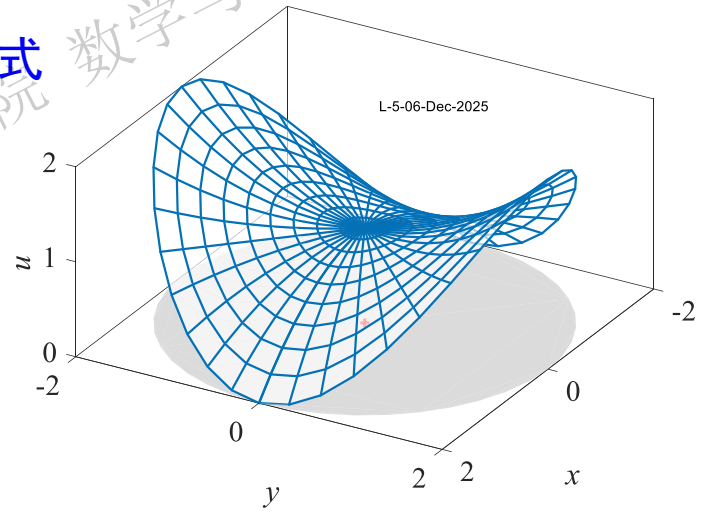
若自由项 $f = 0$, 解转化成圆域上调和函数的Poisson公式

$$u(\rho_0, \theta_0) = \frac{1}{2\pi} \cdot \int_0^{2\pi} \frac{(R^2 - \rho_0^2)\varphi(\theta)}{R^2 + \rho_0^2 - 2\rho_0 R \cos(\theta_0 - \theta)} d\theta$$



S.Poisson
1781-1840
French Math.
Phys.

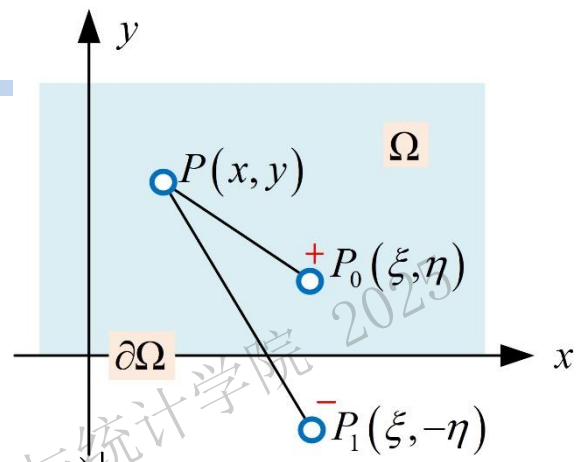
若圆周满足边界条件 $\varphi(\theta) = 2 \sin^2 \theta$



- 1. 镜像法不仅可以调整对称点位置, 还可以调整电荷电量
- 2. 历史上, 本例的证明使得Green函数法得到了广泛认可
- 3. 第2章Poisson公式的推导→Green函数亦可通过解析法得到
- 4. Green函数积分形式解是Poisson公式的推广→自由项的影响

5.4 半空间/圆域上的Dirichlet问题

例3 利用Green函数法求解
$$\begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, & -\infty < x < +\infty, 0 < y \\ u = \varphi(x), & -\infty < x < +\infty, y = 0 \end{cases}$$



解 借助Green函数, 定解问题的通解为

$$u(\xi, \eta) = - \int_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iint_{\Omega} G f \cdot dV$$

二维

关键问题 在题设区域下构造 $G(P, P_0)$

使其满足
$$\begin{cases} -\Delta G = \delta(P, P_0), & P(x, y) \in \Omega \\ G(P, P_0) = 0, & P(x, y) \in \partial\Omega \end{cases}$$

为保证边界上电位为0→根据镜像法

$$\begin{aligned} G(P, P_0) &= \Gamma(P, P_0) - \Gamma(P, P_1) \\ &= \frac{1}{2\pi} \ln \frac{1}{r_{PP_0}} - \frac{1}{2\pi} \ln \frac{1}{r_{PP_1}} = \frac{1}{2\pi} \ln \frac{r_{PP_1}}{r_{PP_0}} \\ &= \frac{1}{4\pi} \ln \frac{(x-\xi)^2 + (y-(-\eta))^2}{(x-\xi)^2 + (y-\eta)^2} \end{aligned}$$

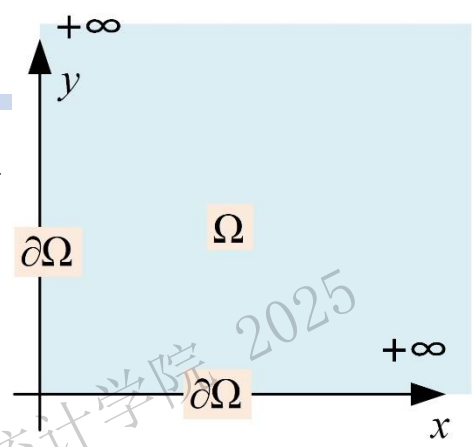
$$\begin{aligned} \frac{\partial G(P, P_0)}{\partial n} \Big|_{y=0} &= \frac{\partial G(P, P_0)}{-\partial y} \Big|_{y=0} \\ &= -\frac{1}{4\pi} \left[\frac{2(y+\eta)}{(x-\xi)^2 + (y+\eta)^2} - \frac{2(y-\eta)}{(x-\xi)^2 + (y-\eta)^2} \right]_{y=0} \\ &= -\frac{1}{\pi} \frac{\eta}{(x-\xi)^2 + \eta^2} \end{aligned}$$

原定解问题的解可表示为

$$\begin{aligned} u(\xi, \eta) &= - \int_{\partial\Omega} \varphi \frac{\partial G}{\partial n} ds + \iint_{\Omega} G \cdot 0 \cdot dV \\ &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{\varphi(x) \eta}{(x-\xi)^2 + \eta^2} dx \end{aligned}$$

5.4 半空间/圆域上的Dirichlet问题

例4 以平面直角坐标系第一象限为研究区域 $\Omega = \{(x, y) \mid x \geq 0, y \geq 0\}$
其边界为 $\partial\Omega = \{(x, y) \mid x = 0, y \geq 0\} \cup \{(x, y) \mid y = 0, x \geq 0\}$,
请利用镜像法求出该区域的Green函数



解 该区域上的Green函数是下述定解问题的解

$$\begin{cases} -\Delta G = \delta(P, P_0), & P(x, y) \in \Omega \\ G(P, P_0) = 0, & P(x, y) \in \partial\Omega \end{cases}$$

第一象限内任意点 $P_0(\xi, \eta)$ 放置单位**正**电荷

则基本解 $\Gamma(P, P_0) = \frac{1}{2\pi} \ln \frac{1}{r_{P_0P}}$ 满足方程

为保证边界上电位为0, 需在

$(-\xi, -\eta)$ 放置单位**正**电荷

$(-\xi, \eta)$ $(\xi, -\eta)$ 放置单位**负**电荷

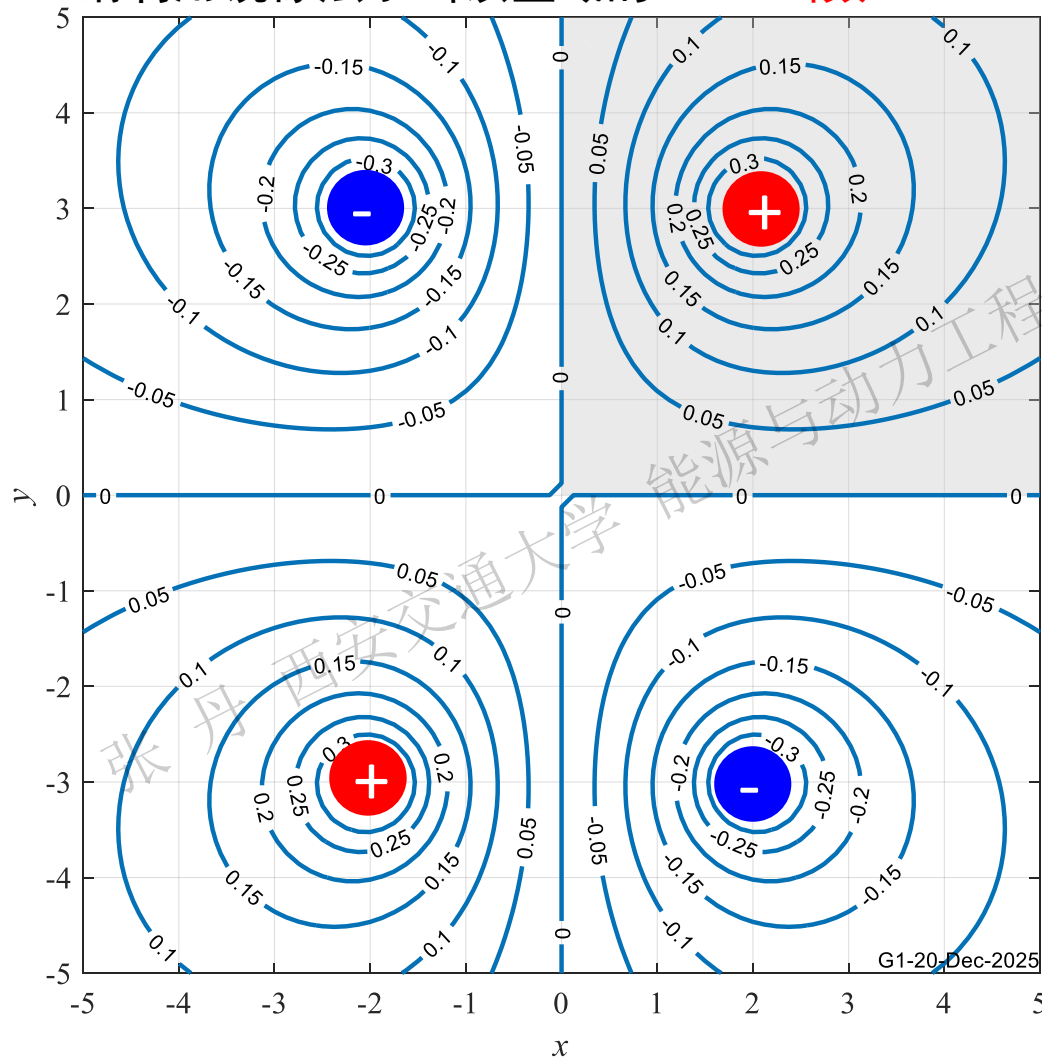
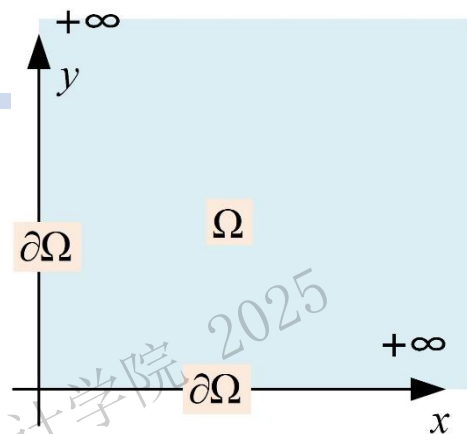
Green函数用4个基本解组合表示

$$\begin{aligned} G(P, P_0) = & \frac{1}{2\pi} \ln \frac{1}{\sqrt{(x-\xi)^2 + (y-\eta)^2}} \\ & - \frac{1}{2\pi} \ln \frac{1}{\sqrt{(x-\xi)^2 + (y+\eta)^2}} \\ & - \frac{1}{2\pi} \ln \frac{1}{\sqrt{(x+\xi)^2 + (y-\eta)^2}} \\ & + \frac{1}{2\pi} \ln \frac{1}{\sqrt{(x+\xi)^2 + (y+\eta)^2}} \end{aligned}$$

$$G(x, y) = \frac{1}{4\pi} \ln \frac{\left[(x-\xi)^2 + (y+\eta)^2 \right] \left[(x+\xi)^2 + (y-\eta)^2 \right]}{\left[(x-\xi)^2 + (y-\eta)^2 \right] \left[(x+\xi)^2 + (y+\eta)^2 \right]}$$

5.4 半空间/圆域上的Dirichlet问题

例4 以平面直角坐标系第一象限为研究区域 $\Omega = \{(x, y) \mid x \geq 0, y \geq 0\}$
其边界为 $\partial\Omega = \{(x, y) \mid x=0, y \geq 0\} \cup \{(x, y) \mid y=0, x \geq 0\}$,
请利用镜像法求出该区域的Green函数



$$G(x, y) = \frac{1}{4\pi} \left[\ln \frac{(x-\xi)^2 + (y+\eta)^2}{(x-\xi)^2 + (y-\eta)^2} + \ln \frac{(x+\xi)^2 + (y-\eta)^2}{(x+\xi)^2 + (y+\eta)^2} \right]$$

5.4 半空间/圆域上的Dirichlet问题

例5 利用镜像法求出区域 $\Omega = \{(x, y, z) \mid z > x + y\}$ 上的Green函数

解 设 $P_0(\xi, \eta, \zeta) \in \Omega$, 寻找关于边界的

对称点 $P_1(\xi_1, \eta_1, \zeta_1) \in \Omega$

题设半空间边界为斜平面

$$F = x + y - z = 0$$

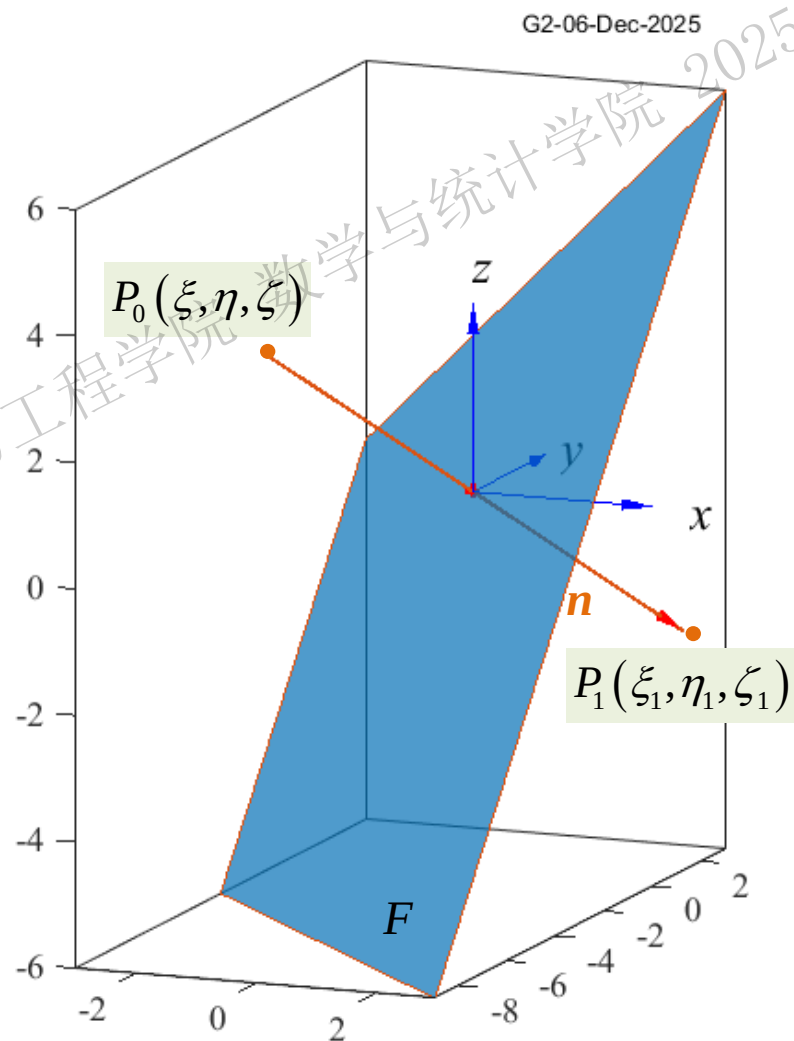
该平面的外法线为 $\mathbf{n} = \nabla F = (1, 1, -1)$

连线 P_0P_1 与 $\mathbf{n}$ 平行

$$P_0P_1 \times \mathbf{n} = \mathbf{0} = \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & -1 \\ \xi_1 - \xi & \eta_1 - \eta & \zeta_1 - \zeta \end{bmatrix}$$

得到连线方程的参数形式

$$\frac{\xi_1 - \xi}{1} = \frac{\eta_1 - \eta}{1} = \frac{\zeta_1 - \zeta}{-1} = t$$



5.4 半空间/圆域上的Dirichlet问题

例5 利用镜像法求出区域 $\Omega = \{(x, y, z) \mid z > x + y\}$ 上的Green函数

G2-06-Dec-2025

两点连线的中点坐标为

$$\left. \begin{aligned} \frac{\xi_1 + \xi}{2} &= \xi + \frac{\xi_1 - \xi}{2} = \xi + \frac{t}{2} \\ \frac{\eta_1 + \eta}{2} &= \eta + \frac{\eta_1 - \eta}{2} = \eta + \frac{t}{2} \\ \frac{\zeta_1 + \zeta}{2} &= \zeta + \frac{\zeta_1 - \zeta}{2} = \zeta - \frac{t}{2} \end{aligned} \right\}$$

中点坐标满足
边界面 F 方程

$$F = x + y - z = 0$$
$$t = \frac{-2(\xi + \eta - \zeta)}{3}$$

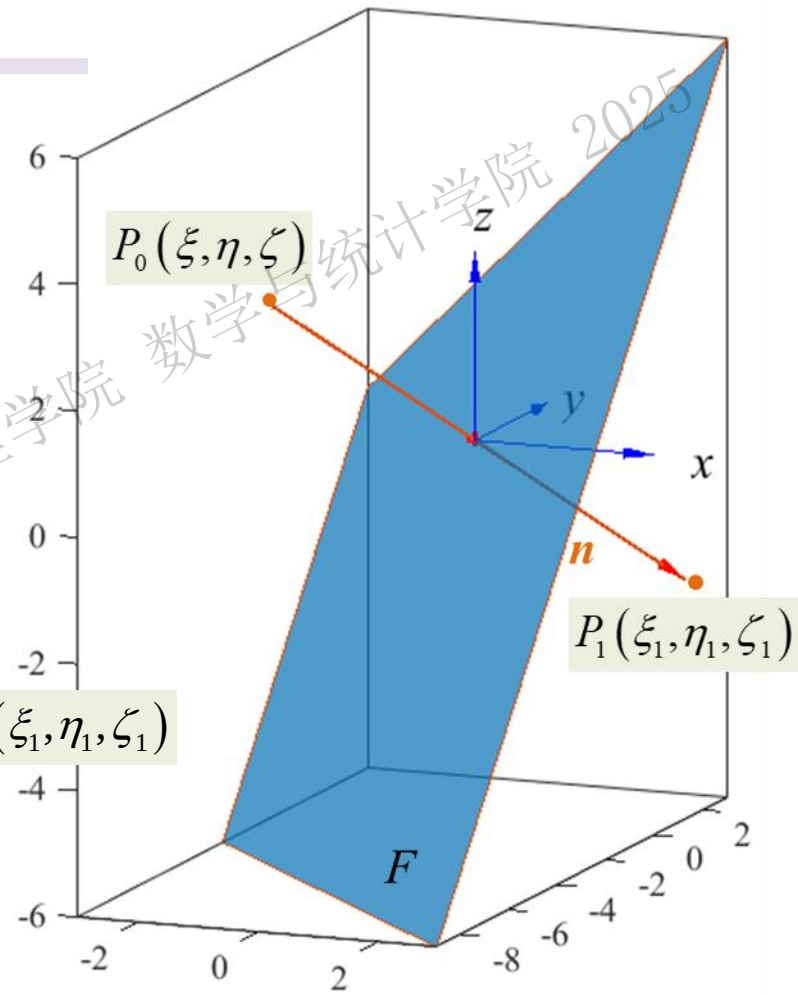
对称点的坐标

$$\left\{ \begin{aligned} \xi_1 &= \xi + t = \xi + \frac{-2(\xi + \eta - \zeta)}{3} = \frac{\xi - 2\eta + 2\zeta}{3} \\ \eta_1 &= \eta + t = \eta + \frac{-2(\xi + \eta - \zeta)}{3} = \frac{-2\xi + \eta + 2\zeta}{3} \\ \zeta_1 &= \zeta - t = \zeta + \frac{2(\xi + \eta - \zeta)}{3} = \frac{2\xi + 2\eta + \zeta}{3} \end{aligned} \right\}$$

设点 $P(x, y, z) \in \Omega$

则该区域上的Green函数可表示为

$$G(P, P_0) = \frac{1}{4\pi} \left(\frac{1}{|PP_0|} - \frac{1}{|PP_1|} \right)$$



5.4 半空间/圆域上的Dirichlet问题

例6 设 $\Omega \subset R^3$ 为有界区域, $\partial\Omega$ 充分光滑, 考虑定解问题

$$\begin{cases} -\Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ \left. \frac{\partial u(x, y, z)}{\partial n} \right|_{\partial\Omega} = \varphi(x, y, z) & (x, y, z) \in \partial\Omega \end{cases}$$

证明该问题可解的必要条件是 $\iiint_{\Omega} f dV = \iint_{\partial\Omega} \varphi ds$

证 在Green第2公式中令 $v=1$

$$\iiint_{\Omega} (u \Delta v - v \Delta u) dV = \iint_{\partial\Omega} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds$$

$$\iiint_{\Omega} (u \cdot 0 - 1 \cdot \Delta u) dV = \iint_{\partial\Omega} \left(u \cdot 0 - 1 \cdot \frac{\partial u}{\partial n} \right) ds$$

$$\iiint_{\Omega} (-\Delta u) dV = \iint_{\partial\Omega} \left(-\frac{\partial u}{\partial n} \right) ds$$

$$\iiint_{\Omega} f dV = \iint_{\partial\Omega} \varphi ds$$

Green公式

$$\begin{aligned} \iiint_{\Omega} u \Delta v dV &= \iint_{\partial\Omega} u \frac{\partial v}{\partial n} ds - \iiint_{\Omega} \nabla u \cdot \nabla v dV \\ \iiint_{\Omega} (u \Delta v - v \Delta u) dV &= \iint_{\partial\Omega} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds \\ u &= \iint_{\partial\Omega} \left(\Gamma \frac{\partial u}{\partial n} - u \frac{\partial \Gamma}{\partial n} \right) ds - \iiint_{\Omega} \Gamma \Delta u dV \end{aligned}$$

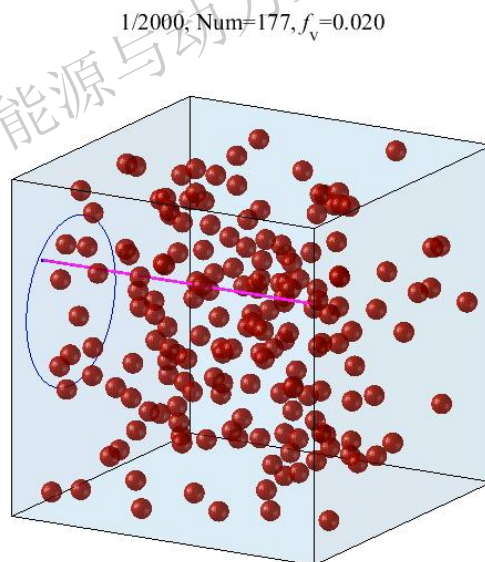
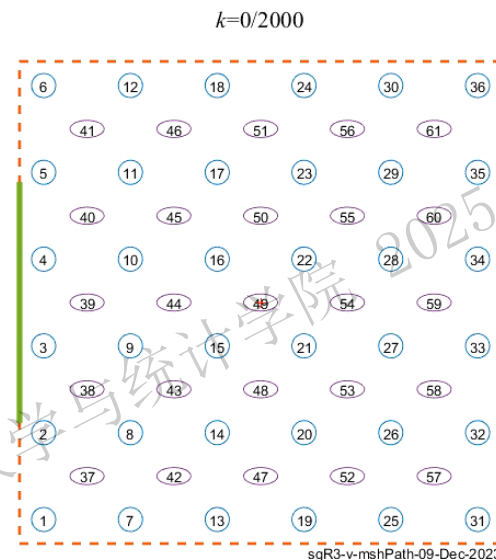
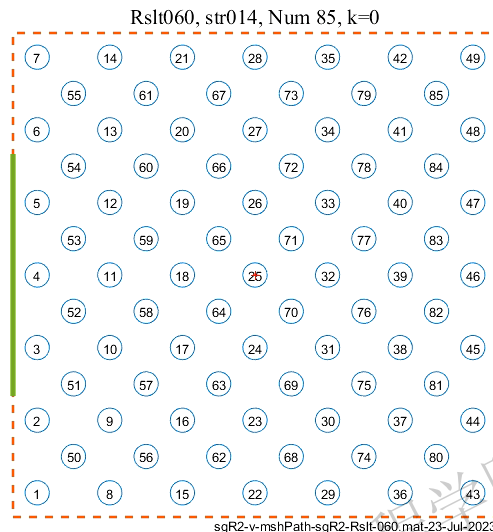
Green函数法启示-数学方程→物理求解

Monte-Carlo Method

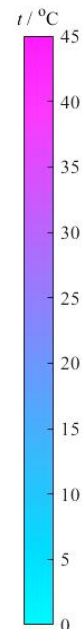
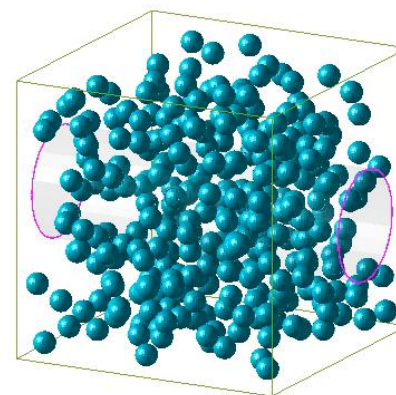
若物理现象控制方程过于复杂→回到物理现象本身去解决→研究复杂物理过程的有效途径



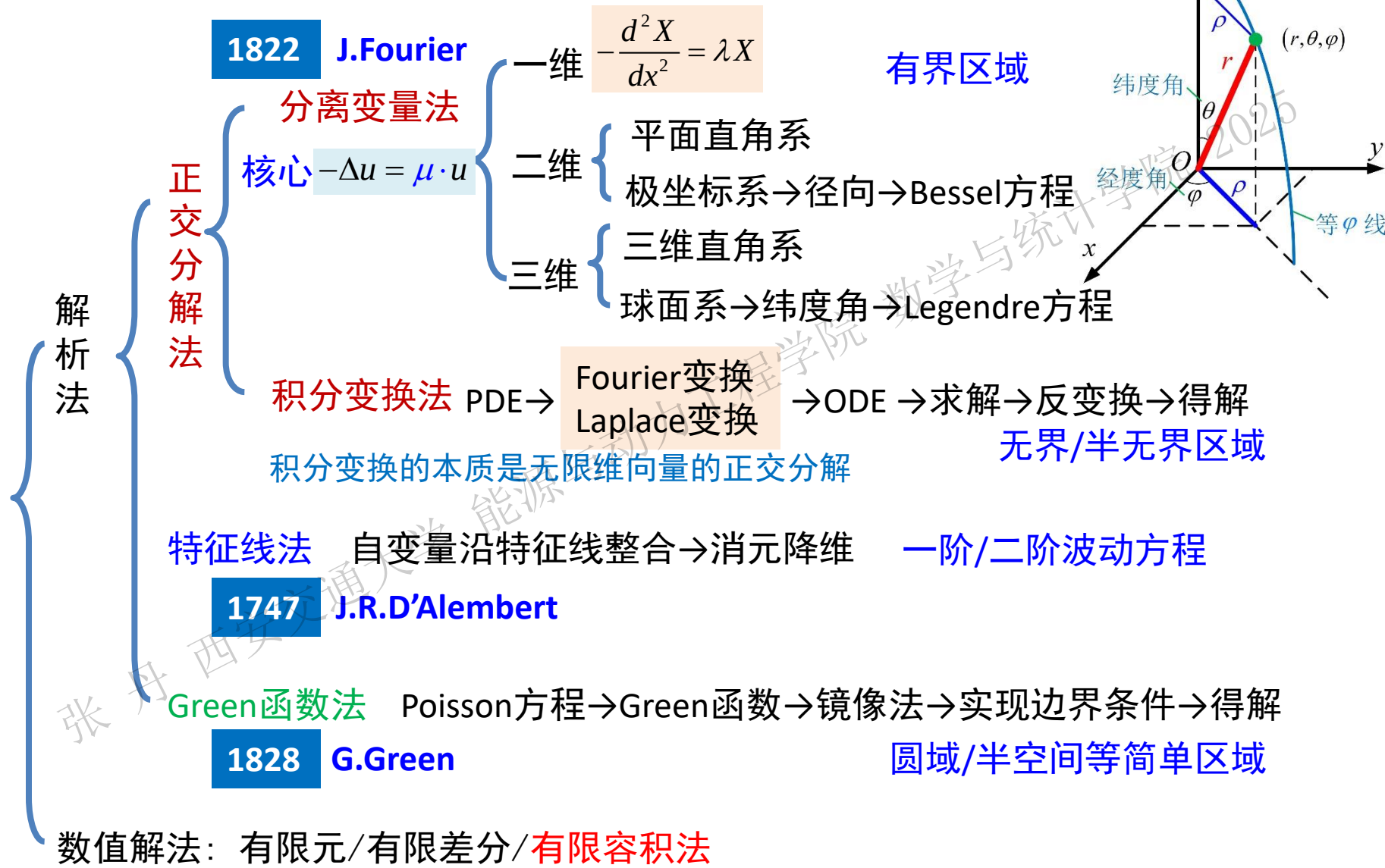
- 太阳能光热开发的核心是粒子系的辐射传热
- 粒子间辐射传热是4重积分→非常困难→用随机光线模拟替代→还原物理意义+概率→解PDE定解问题



$\tau=0.0s$, rStr-241, $D=0.8mm$, Num=299, $f_{v0}=0.08$,



PDE的解法



PDE的解法

经典PDE定解问题的解法

	波动方程	导热方程	Poisson方程	一阶PDE
	$\frac{\partial^2 u}{\partial t^2} = a^2 \Delta u + f$	$\frac{\partial u}{\partial t} = a^2 \Delta u + f$	$0 = \Delta u + f$	$au_x + bu_t = c$
分离变量法	✓	✓	✓	✗
积分变换法	✓	✓	✓	✓
特征线法	✓	✗	✗	✓
Green函数法	✓	✓	✓	✗

本章作业

① 利用Green函数法求解定解问题

$$\begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, & -\infty < x < +\infty, 0 < y \\ u = \varphi(x), & -\infty < x < +\infty, y = 0 \end{cases}$$

$$\begin{cases} -\Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ u(x, y, 0) = \varphi(x, y) & (x, y) \in \partial\Omega \end{cases}$$

其中 $\Omega = \{(x, y, z) \mid z > 0\}$, $\partial\Omega = \{(x, y, z) \mid z = 0\}$

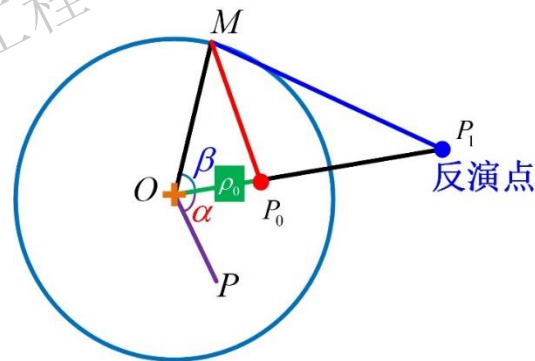
② 利用镜像法求出下述区域上的Green函数

(2.1) 第一象限 $\Omega = \{(x, y) \mid x \geq 0, y \geq 0\}$ $\partial\Omega = \{(x, y) \mid x = 0, y \geq 0\} \cup \{(x, y) \mid y = 0, x \geq 0\}$

(2.2) 证明圆域上的Green函数可表示为

$$G(P, P_0) = \frac{1}{2\pi} \left[\ln \left(\frac{1}{r_{P_0 P}} \right) - \ln \left(\frac{R}{|OP_0|} \frac{1}{r_{P_1 P}} \right) \right]$$

(2.3) 半空间 $\Omega = \{(x, y, z) \mid z > x + y\}$



③ 证明 设 $\Omega \subset R^3$ 为有界区域, $\partial\Omega$ 充分光滑, 证明下述定解问题可解的必要条件是

$$\begin{cases} \Delta u = f(x, y, z), & (x, y, z) \in \Omega \\ \left. \frac{\partial u(x, y, z)}{\partial n} \right|_{\partial\Omega} = \varphi(x, y, z) & (x, y, z) \in \partial\Omega \end{cases}$$

$$\iiint_{\Omega} f dV = \iint_{\partial\Omega} \varphi ds$$